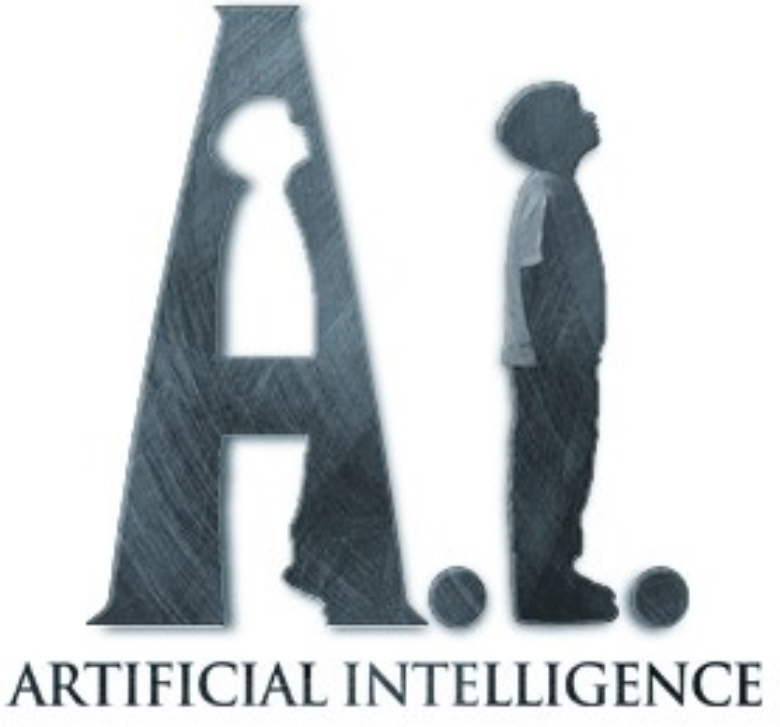
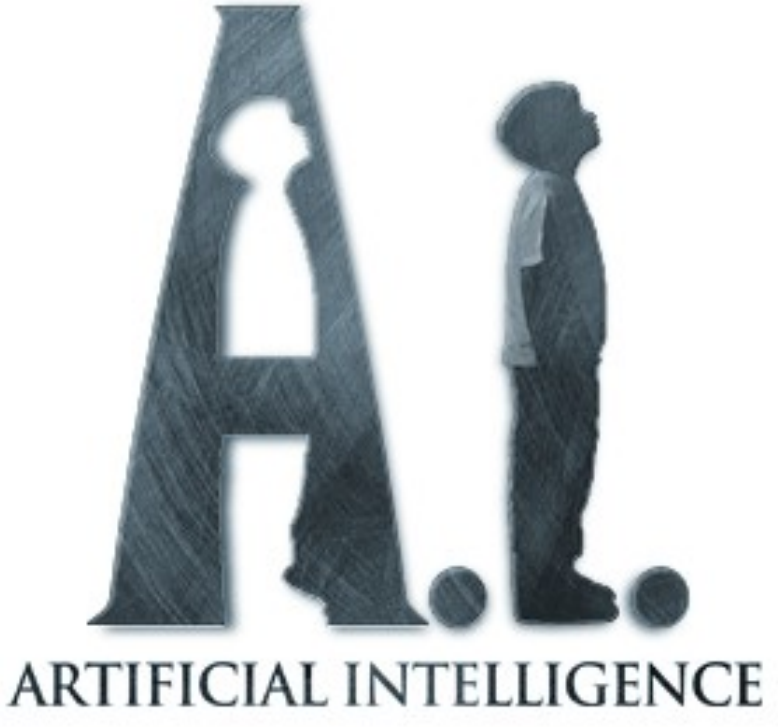


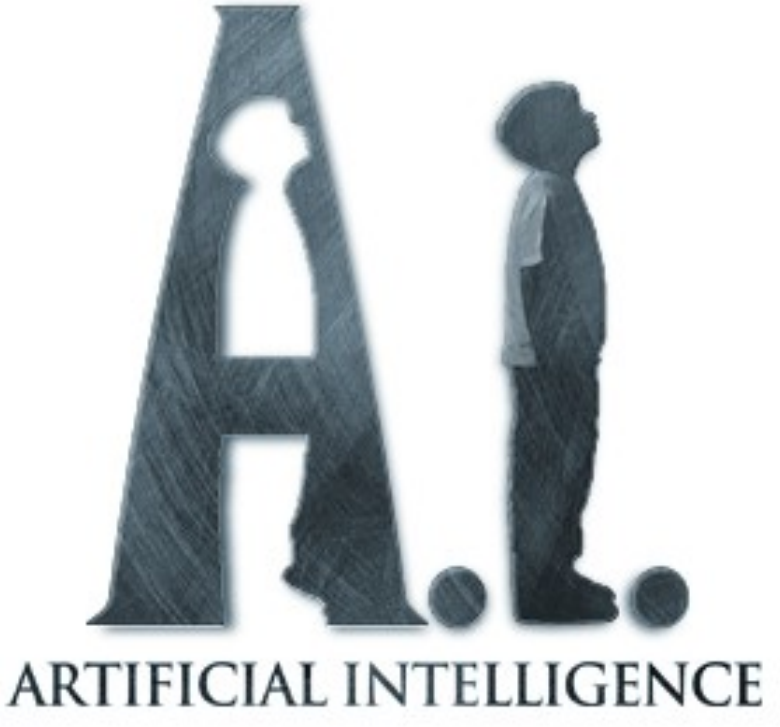
SDAI, something about the course

Viviana Mascardi, viviana.mascardi@unige.it



SDAI, something about the course

https://2026.aulaweb.unige.it/course/view.php?id=1775



IA: a short introduction

Viviana Mascardi, viviana.mascardi@unige.it

What is intelligence?

Intelligĕre: "understand", “read inside", "read beyond the surface", "read in between"

What is intelligence?

Intelligence is a very general mental capability that, among other things, involves the ability to reason, plan, solve problems, think abstractly, comprehend complex ideas, learn quickly, and learn from experience. It is not merely book-learning, a narrow academic skill, or test-taking smarts. Rather, it reflects a broader and deeper capability for comprehending our surroundings "catching on," "making sense" of things, or "figuring out" what to do.

Mainstream Science on Intelligence, Wall Street Journal, Dec, 1994.
Signed by 52 scientists and containing statements I do not feel comfortable with...!

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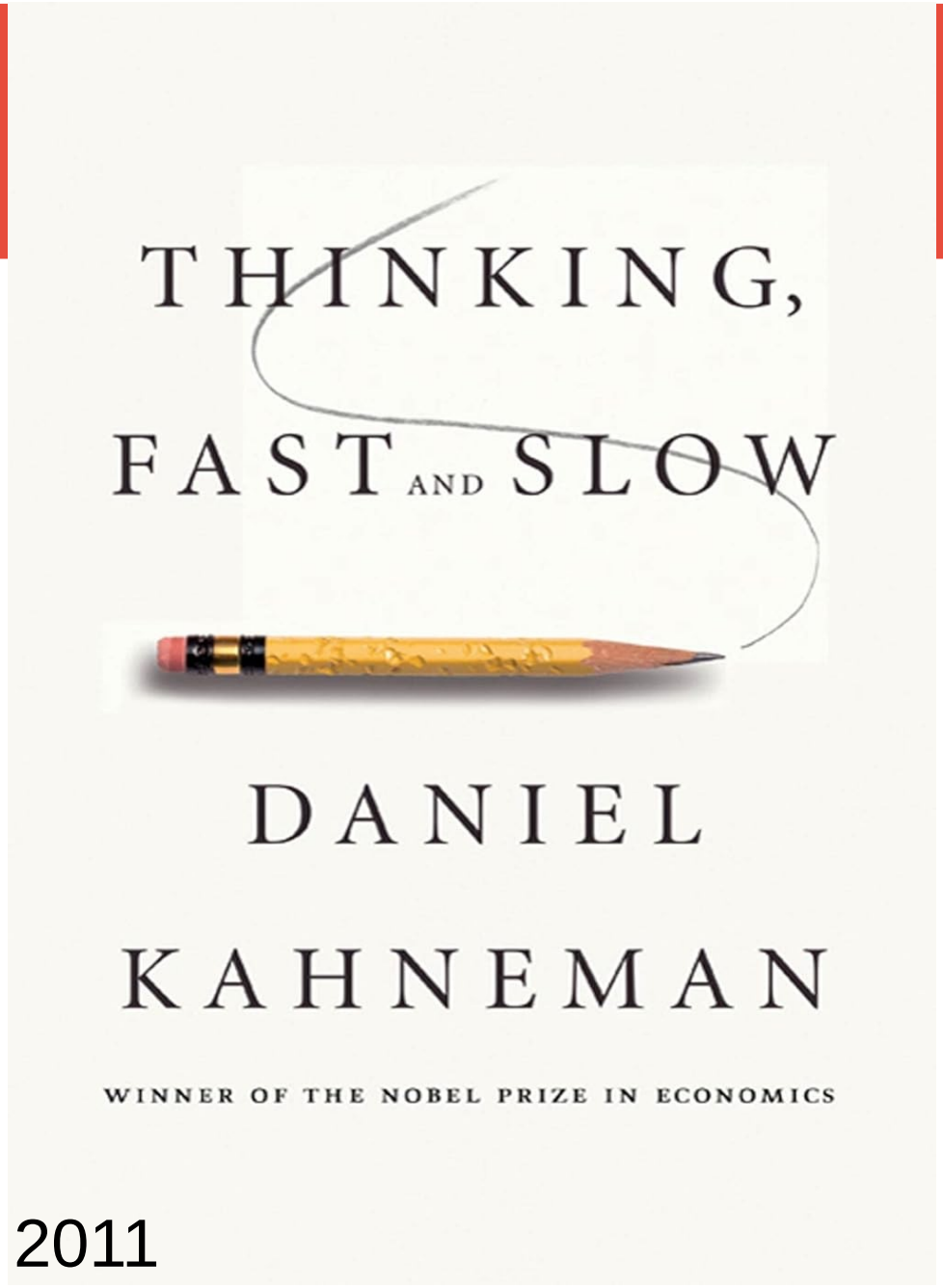
1 Musical-rhythmic and harmonic
2 Visual-spatial
3 Verbal-linguistic
4 Logical-mathematical
5 Bodily-kinesthetic
6 Interpersonal
7 Intrapersonal
8 Naturalistic
9 Existential

H. Gardner (1983). Frames of Mind:
The theory of multiple intelligences.

Emotional Intelligence

Emotional intelligence (EI) is the capability of individuals to recognize their own emotions and those of others, discern between different feelings and label them appropriately, use emotional information to guide thinking and behavior, and manage and/or adjust emotions to adapt to environments or achieve one's goal(s)

D. Goleman (1995), Emotional Intelligence,
Bantam Books


THINKING,
FAST AND SLOW
DANIEL
KAHNEMAN
WINNER OF THE NOBEL PRIZE IN ECONOMICS

2011

System 1: Fast, automatic, frequent, emotional, stereotypic, unconscious

Examples (in order of complexity) of things system 1 can do:

- determine that an object is at a greater distance than another
- localize the source of a specific sound
- complete the phrase "war and ..."
- display disgust when seeing a gruesome image
- solve 2+2=?
- read text on a billboard
- drive a car on an empty road
- think of a good chess move (if you're a chess master)
- understand simple sentences

Source: Wikipedia

System 2: Slow, effortful, infrequent, logical, calculating, conscious

- prepare yourself for the start of a sprint
- direct your attention towards the clowns at the circus
- look for the woman with the grey hair
- try to recognize a sound
- sustain a faster-than-normal walking rate
- determine the appropriateness of a behavior in a social setting
- count the number of A's in a certain text
- give someone your telephone number
- park into a tight parking space
- determine the price/quality ratio of two washing machines
- determine the validity of a complex logical reasoning

Source: Wikipedia

Artificial Intelligence

Artificial intelligence (AI) is intelligence demonstrated by machines, unlike the natural intelligence displayed by humans and animals. Leading AI textbooks such as “Artificial Intelligence: A Modern Approach”, 4th edition by Stuart Russell and Peter Norvig, https://aima.cs.berkeley.edu/, define the field as the study of "intelligent agents": any device that perceives its environment and takes actions that maximize its chance of successfully achieving its goals

Wikipedia: https://en.wikipedia.org/wiki/Artificial_intelligence

Before



384–322 a.C.

Image credit: fotografia di Eric Gaba, https://it.wikipedia.org/wiki/Aristotele

Before



1646–1716

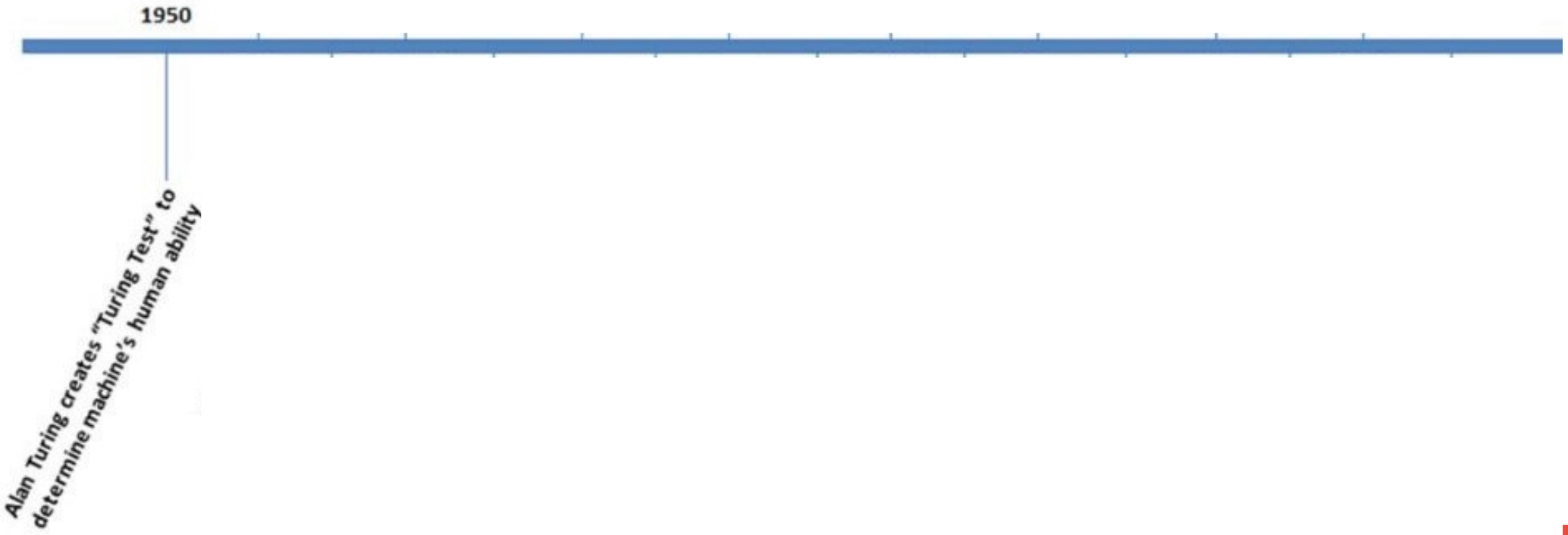
Image credit: Ritratto di Gottfried Wilhelm von Leibniz conservato presso la Biblioteca regionale di Hannover, https://it.wikipedia.org/wiki/Gottfried_Wilhelm_von_Leibniz

Before



1797–1851

Image credit: Ritratto di Mary Shelley (particolare) di Richard Rothwell (1840) esposto alla Royal Academy, https://it.wikipedia.org/wiki/Mary_Shelley


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Alan Turing creates "Turing Test" to determine machine's human ability

VOL. LIX. No. 236.] [October, 1950

MIND

A QUARTERLY REVIEW

OF

PSYCHOLOGY AND PHILOSOPHY

I.—COMPUTING MACHINERY AND INTELLIGENCE

BY A. M. TURING

1. *The Imitation Game.*

I PROPOSE to consider the question, 'Can machines think?' This should begin with definitions of the meaning of the terms 'machine' and 'think'. The definitions might be framed so as to reflect so far as possible the normal use of the words, but this attitude is dangerous. If the meaning of the words 'machine' and 'think' are to be found by examining how they are commonly used it is difficult to escape the conclusion that the meaning and the answer to the question, 'Can machines think?' is to be sought in a statistical survey such as a Gallup poll. But this is absurd. Instead of attempting such a definition I shall replace the question by another, which is closely related to it and is expressed in relatively unambiguous words.

The new form of the problem can be described in terms of a game which we call the 'imitation game'. It is played with three people, a man (A), a woman (B), and an interrogator (C) who may be of either sex. The interrogator stays in a room apart from the other two. The object of the game for the interrogator is to determine which of the other two is the man and which is the woman. He knows them by labels X and Y, and at the end of the game he says either 'X is A and Y is B' or 'X is B and Y is A'. The interrogator is allowed to put questions to A and B thus:

C: Will X please tell me the length of his or her hair?

Now suppose X is actually A, then A must answer. It is A's

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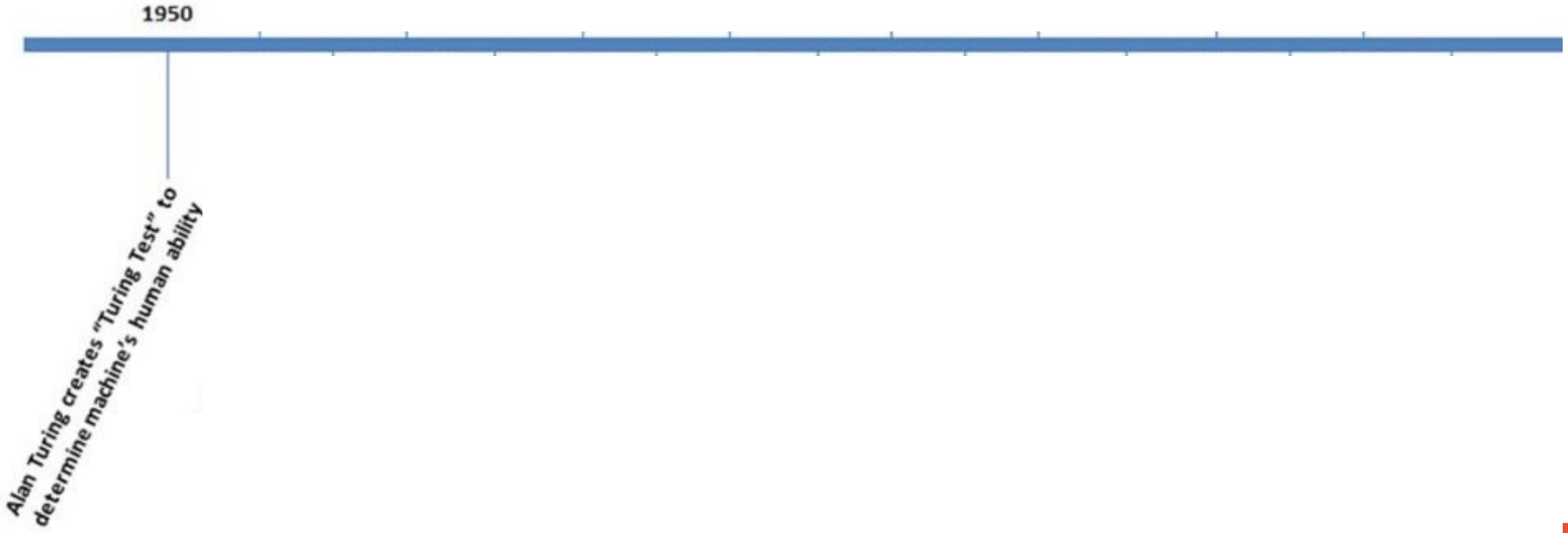
Open parenthesis: what does exacly “think” mean? And to “perceive”? And “to be conscious of being alive”?

https://www.youtube.com/watch?v=Dh5-i2SD1cM

https://www.ted.com/talks/anil_seth_your_brain_hallucinates_your_conscious_reality

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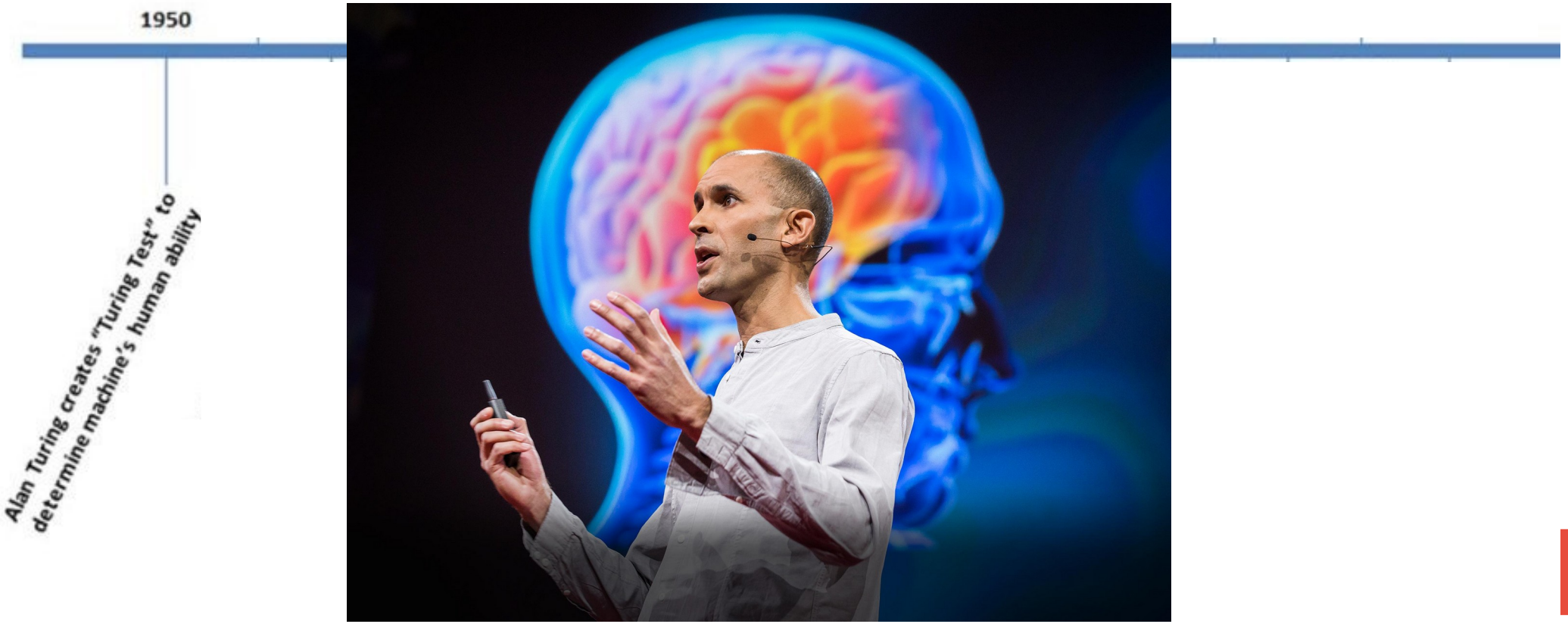
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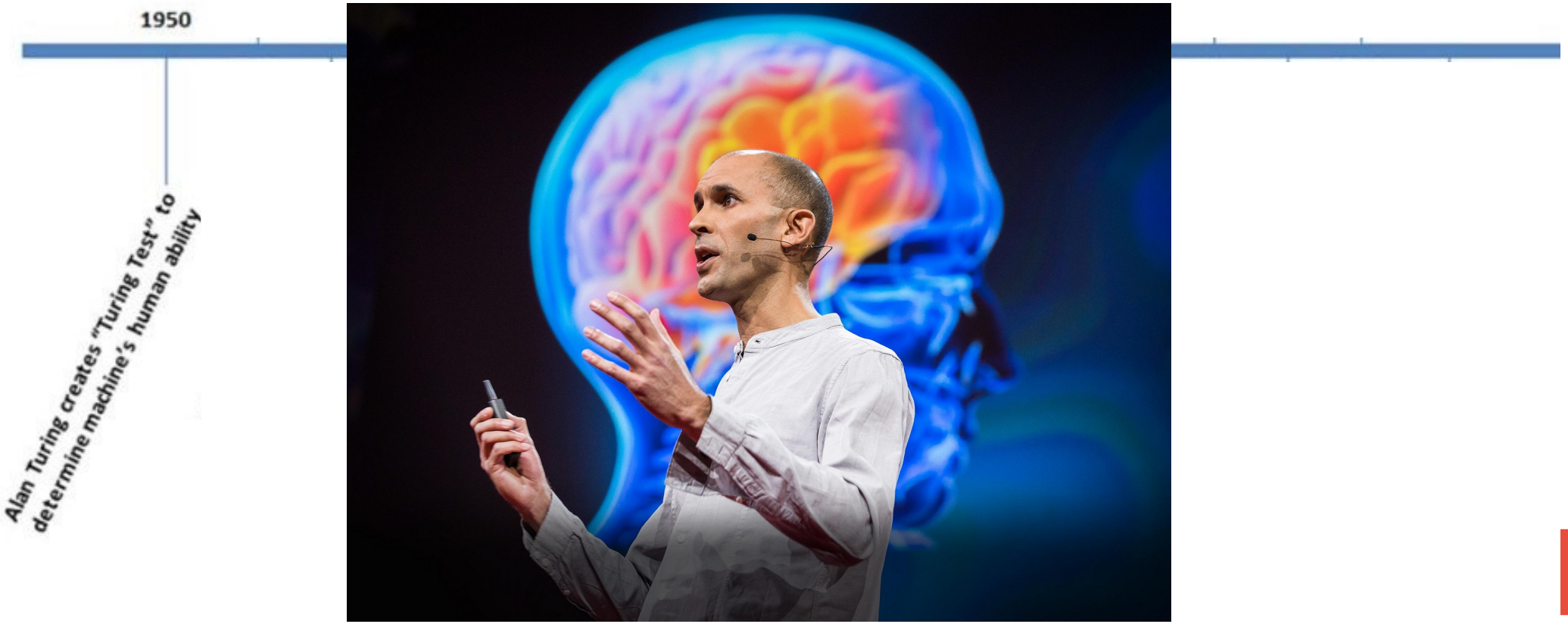
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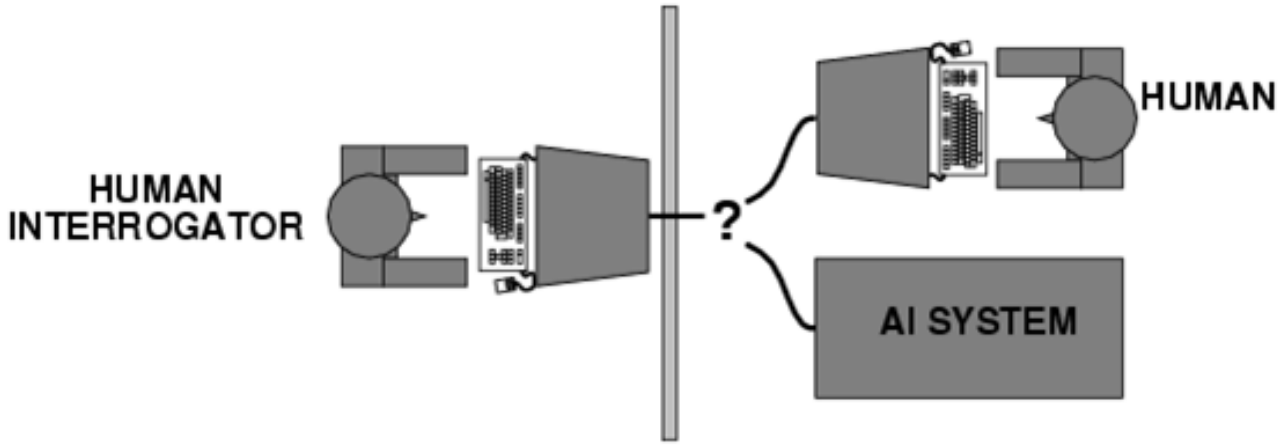
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Closed parenthesis…



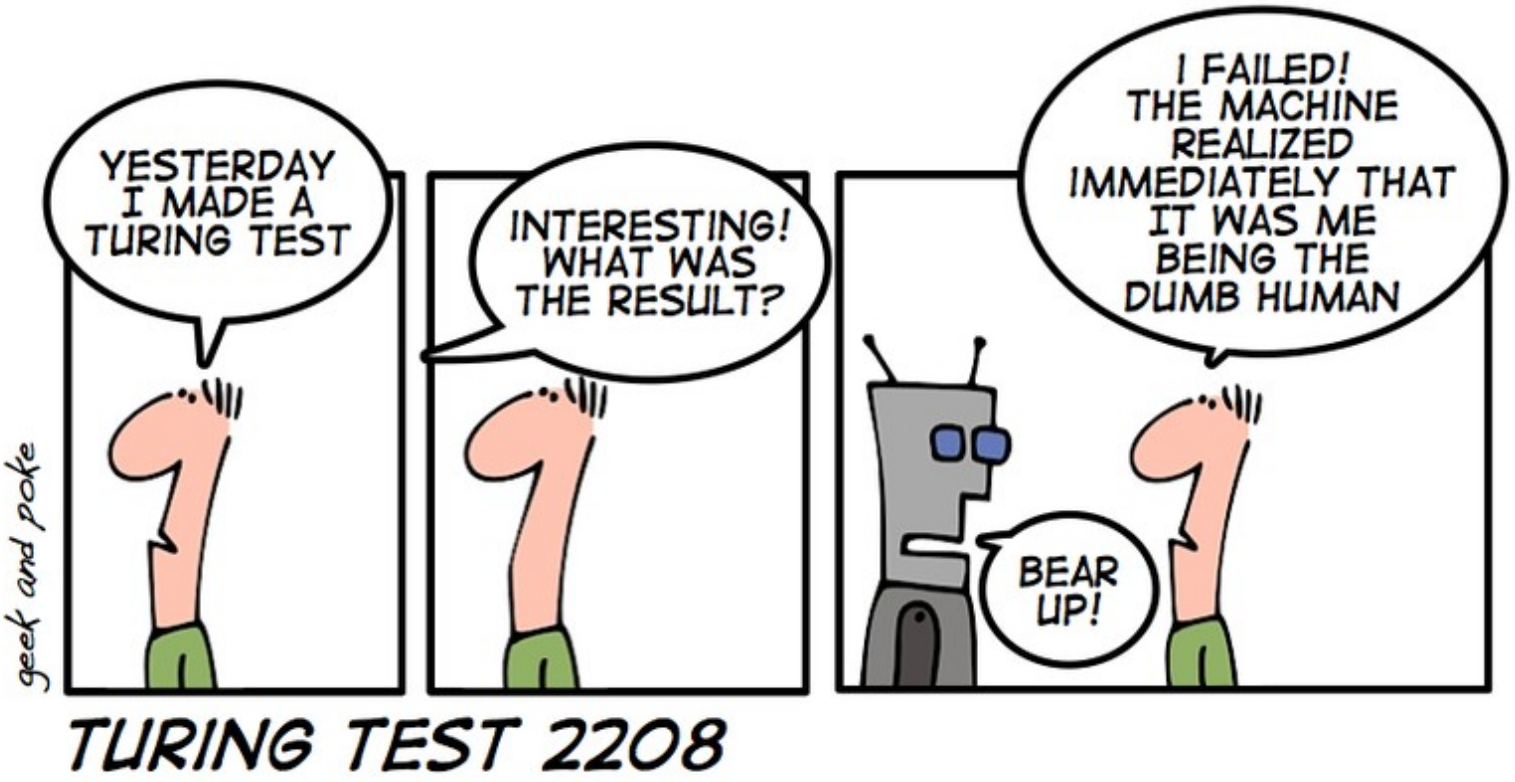


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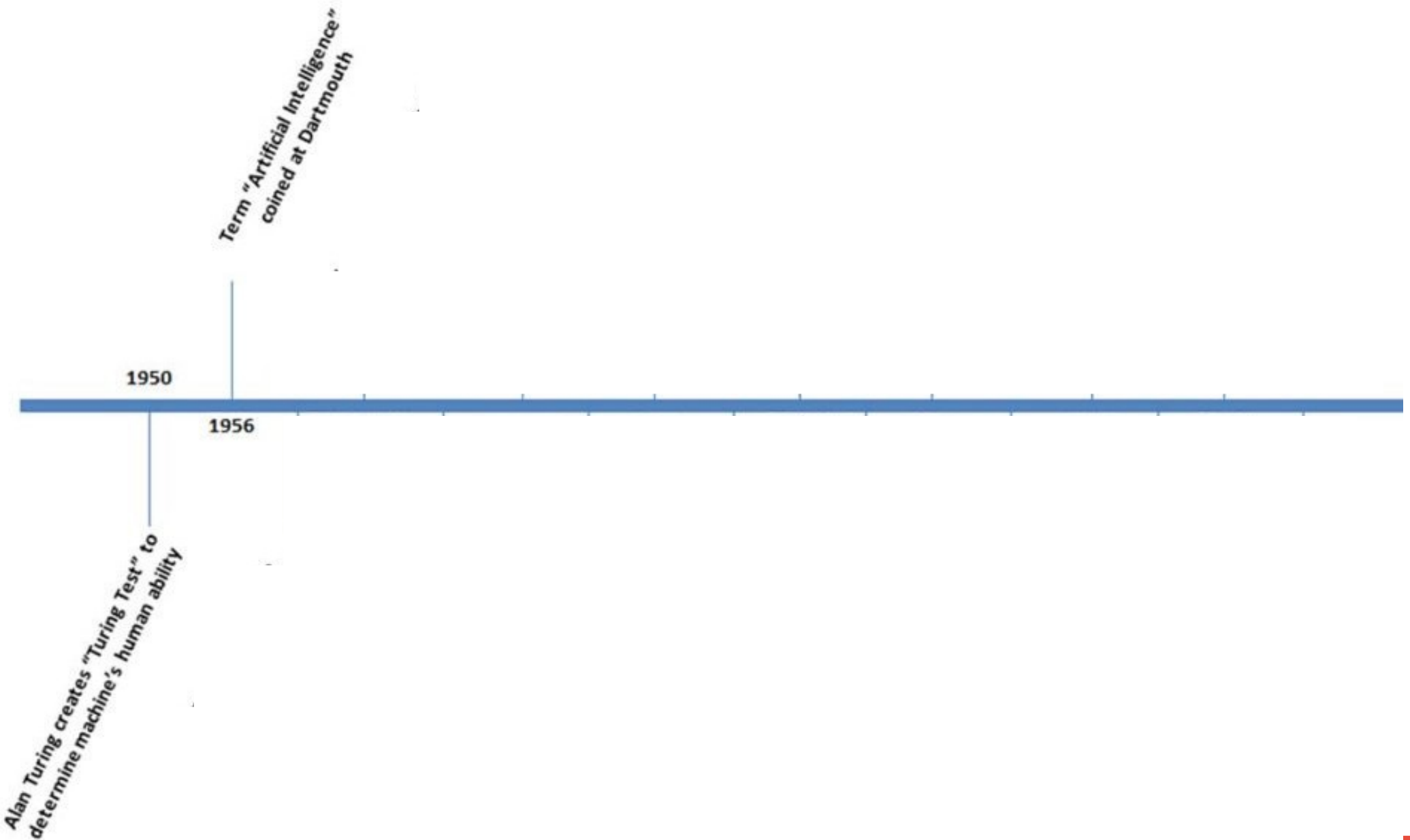
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AI


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Dartmouth Workshop 1956



> "every aspect of learning or any other feature of intelligence can in principle be so precisely described that a machine can be made to simulate it ."

AI

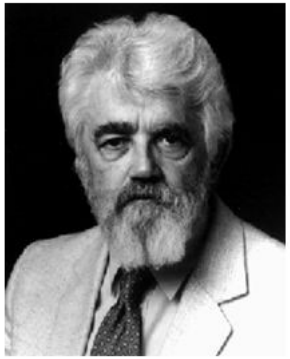
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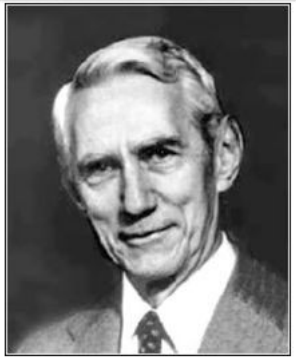
1956 Dartmouth Conference: The Founding Fathers of AI



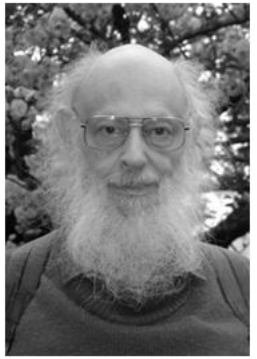
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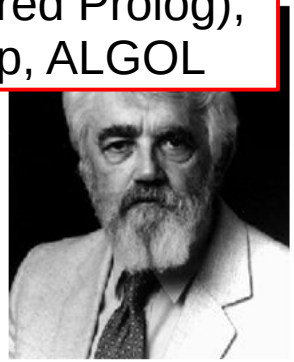
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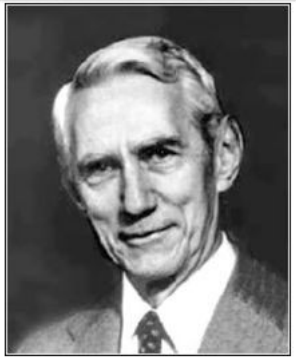
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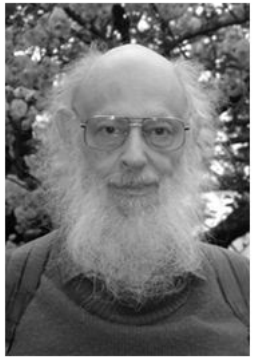
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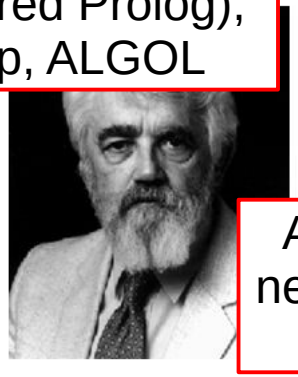
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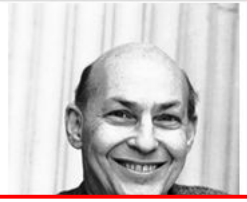
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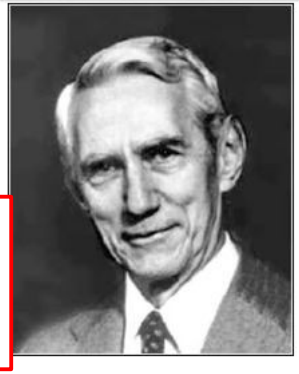


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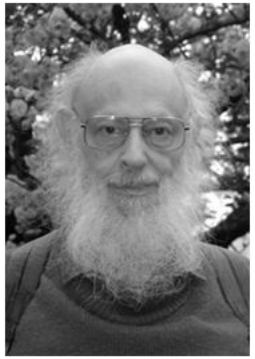


Marvin Minsky

Artificial neural networks, society of mind



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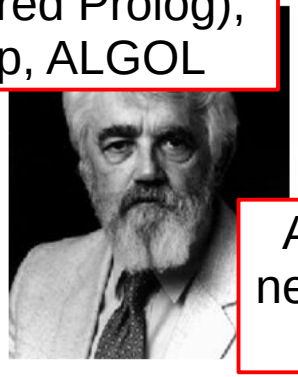
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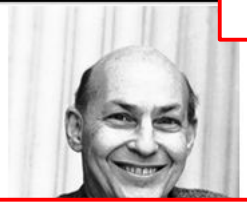
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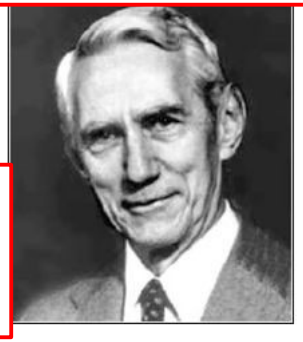
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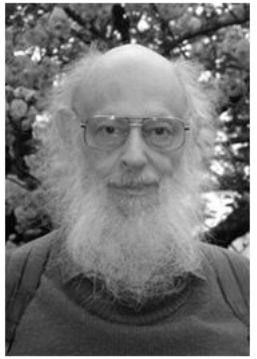
Information theory, cryptography, NLP





Artificial neural networks, society of mind





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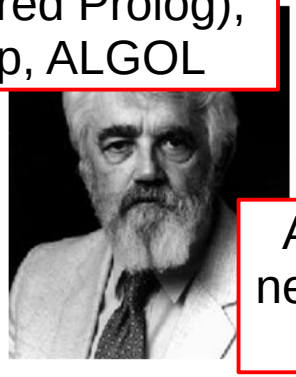
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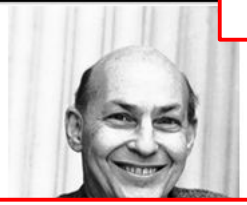
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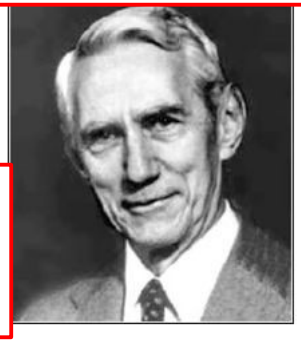
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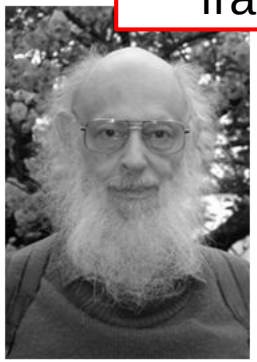
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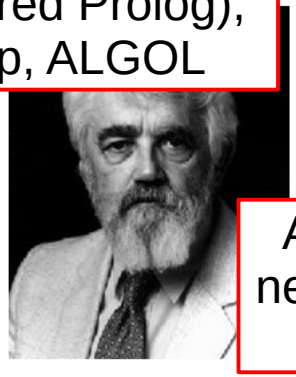
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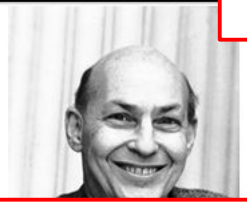
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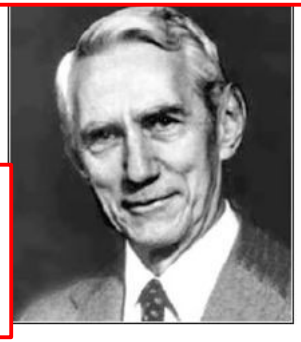
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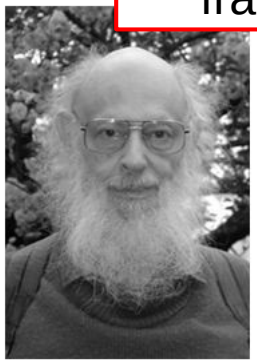
Artificial neural networks, society of mind

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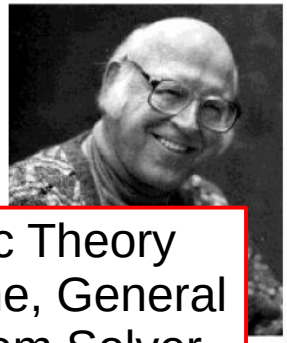
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Logic Theory Machine, General Problem Solver

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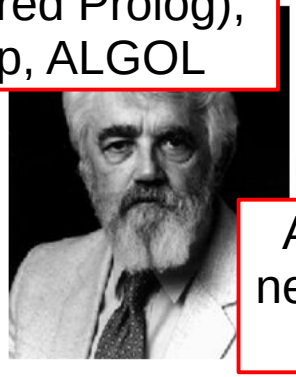
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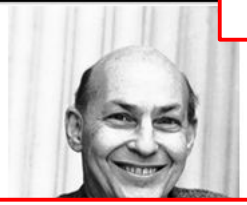
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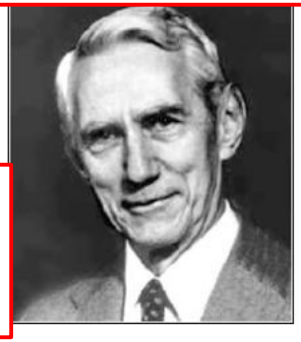
John McCarthy



Marvin Minsky

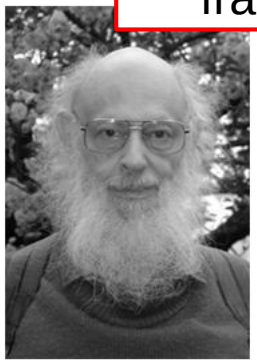
Artificial neural networks, society of mind

Information theory, cryptography, NLP



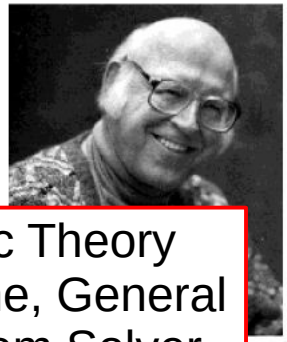
Claude Shannon

ML, prediction, Bayesian frameworks



Ray Solomonoff

Alan Newell



Logic Theory Machine, General Problem Solver

Herbert Simon



Logic Theory Machine, General Problem Solver

Arthur Samuel



And three others…

Oliver Selfridge
(Pandemonium theory)

Nathaniel Rochester
(IBM, designed 701)

Trenchard More
(Natural Deduction)

AI

Term "Artificial Intelligence" coined at Dartmouth

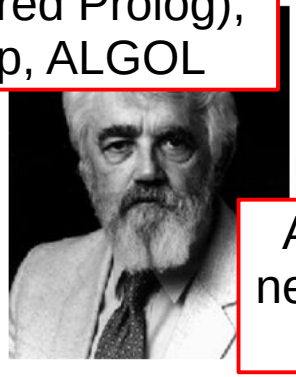
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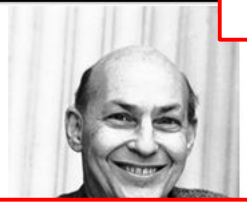
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Advice Taker (inspired Prolog), Lisp, ALGOL

1956 Dartm
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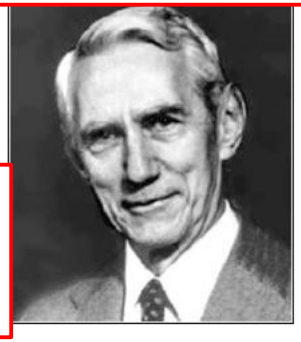
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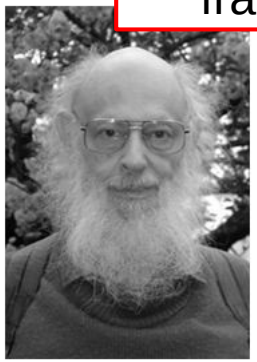
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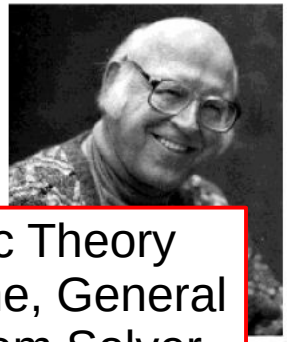
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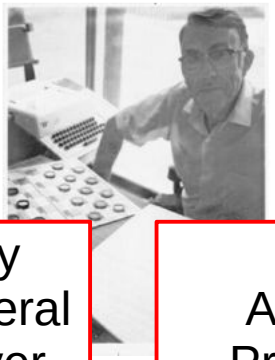
Logic Theory Machine, General Problem Solver

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Search, Alpha-Beta Pruning, ML

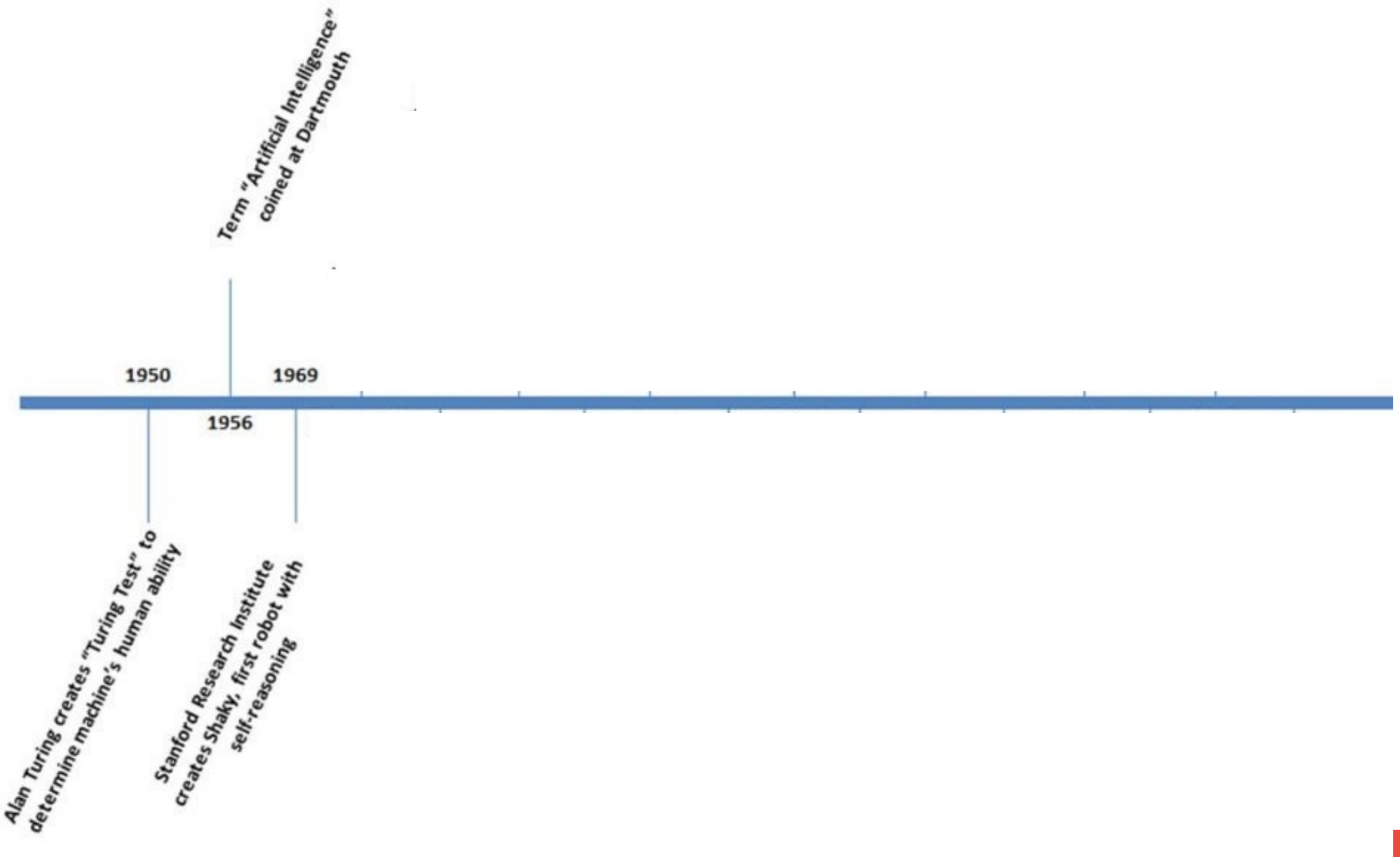
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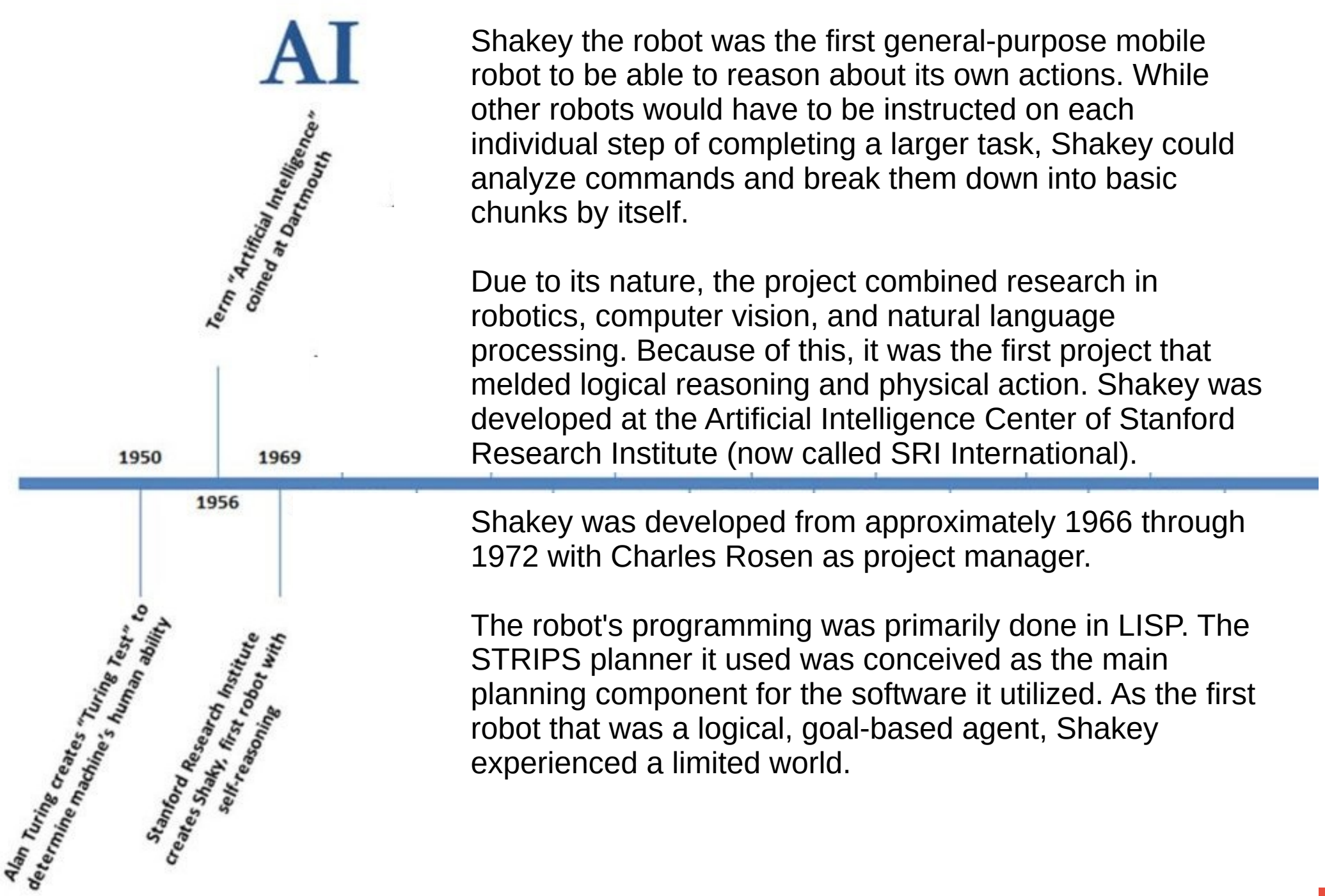
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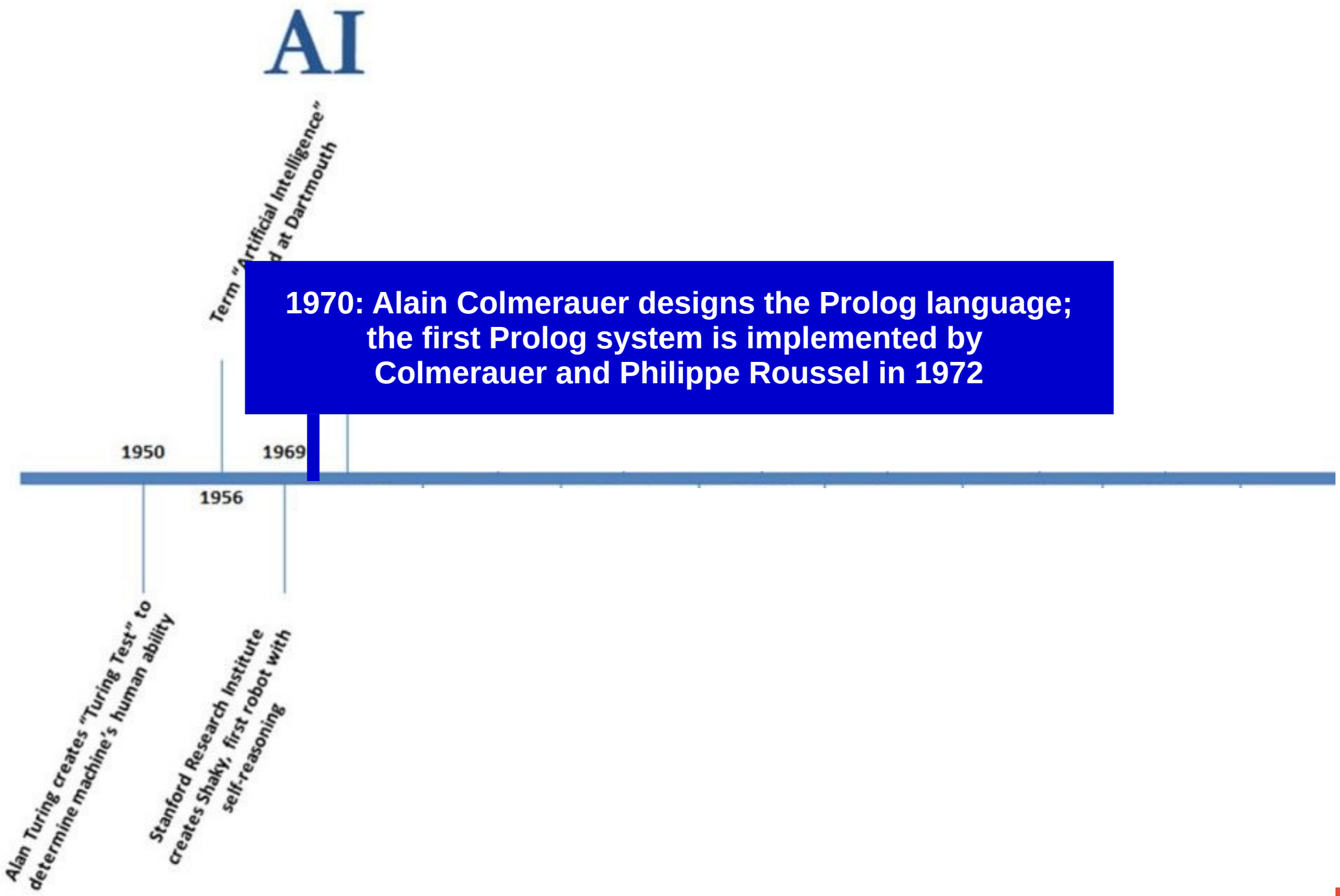


Shakey the robot was the first general-purpose mobile robot to be able to reason about its own actions. While other robots would have to be instructed on each individual step of completing a larger task, Shakey could analyze commands and break them down into basic chunks by itself.

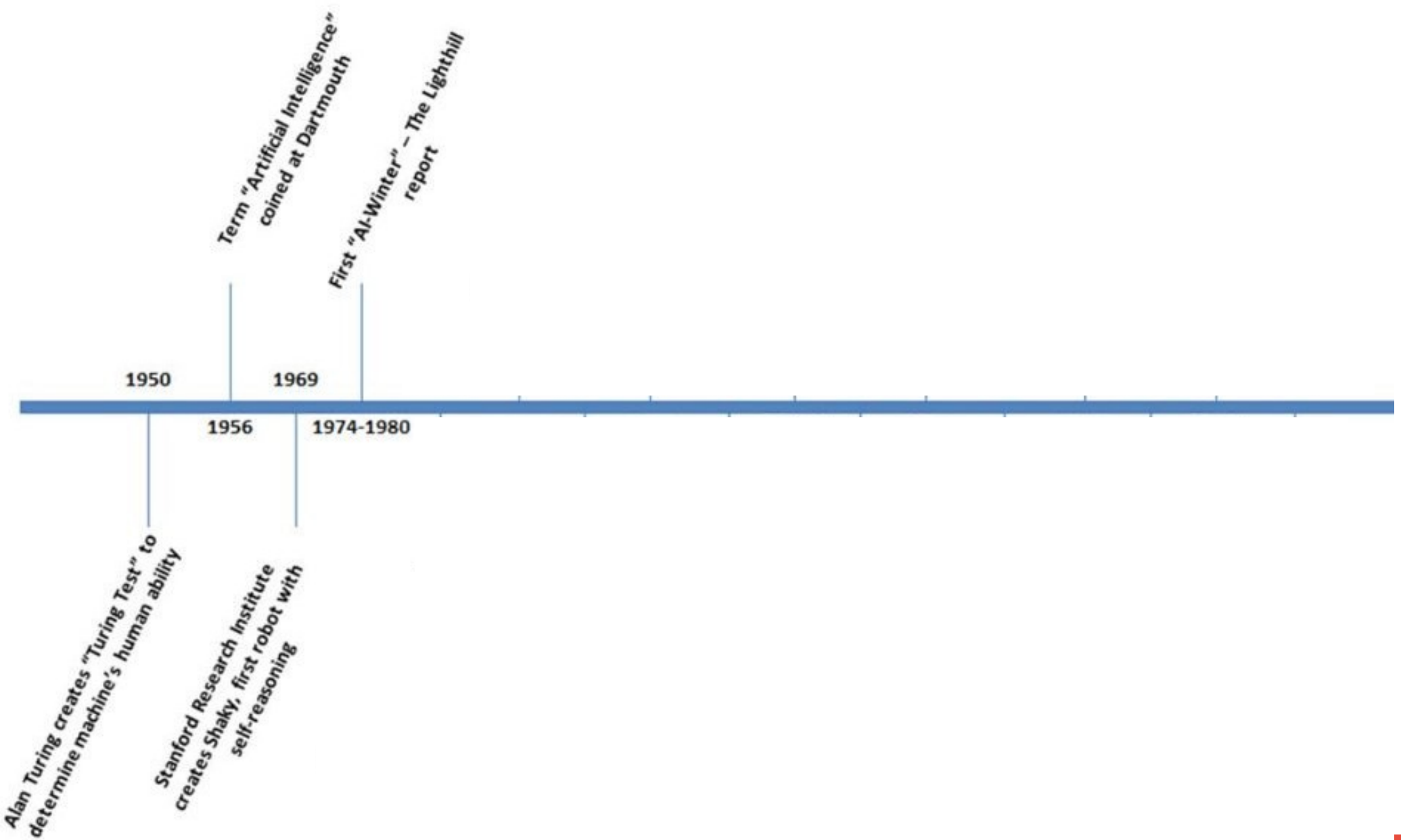
Due to its nature, the project combined research in robotics, computer vision, and natural language processing. Because of this, it was the first project that melded logical reasoning and physical action. Shakey was developed at the Artificial Intelligence Center of Stanford Research Institute (now called SRI International).

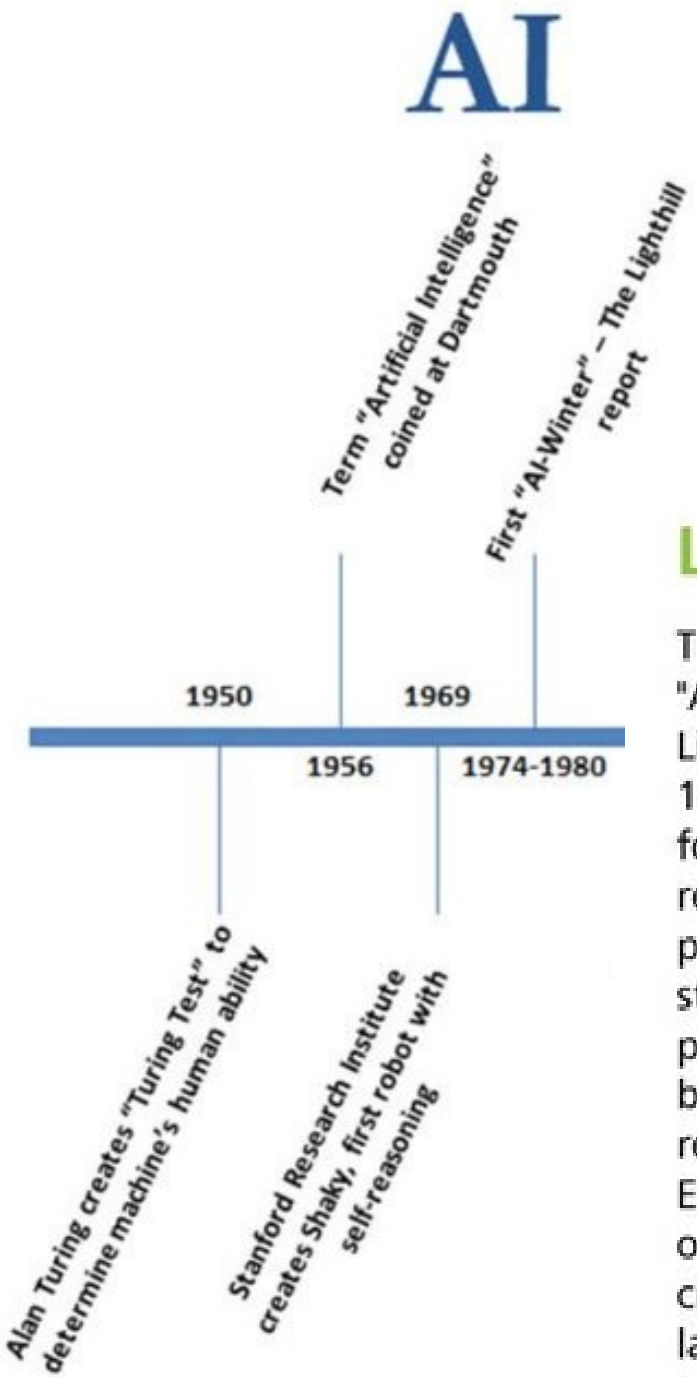
Shakey was developed from approximately 1966 through 1972 with Charles Rosen as project manager.

The robot's programming was primarily done in LISP. The STRIPS planner it used was conceived as the main planning component for the software it utilized. As the first robot that was a logical, goal-based agent, Shakey experienced a limited world.


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1970: Alain Colmerauer designs the Prolog language;
the first Prolog system is implemented by
Colmerauer and Philippe Roussel in 1972
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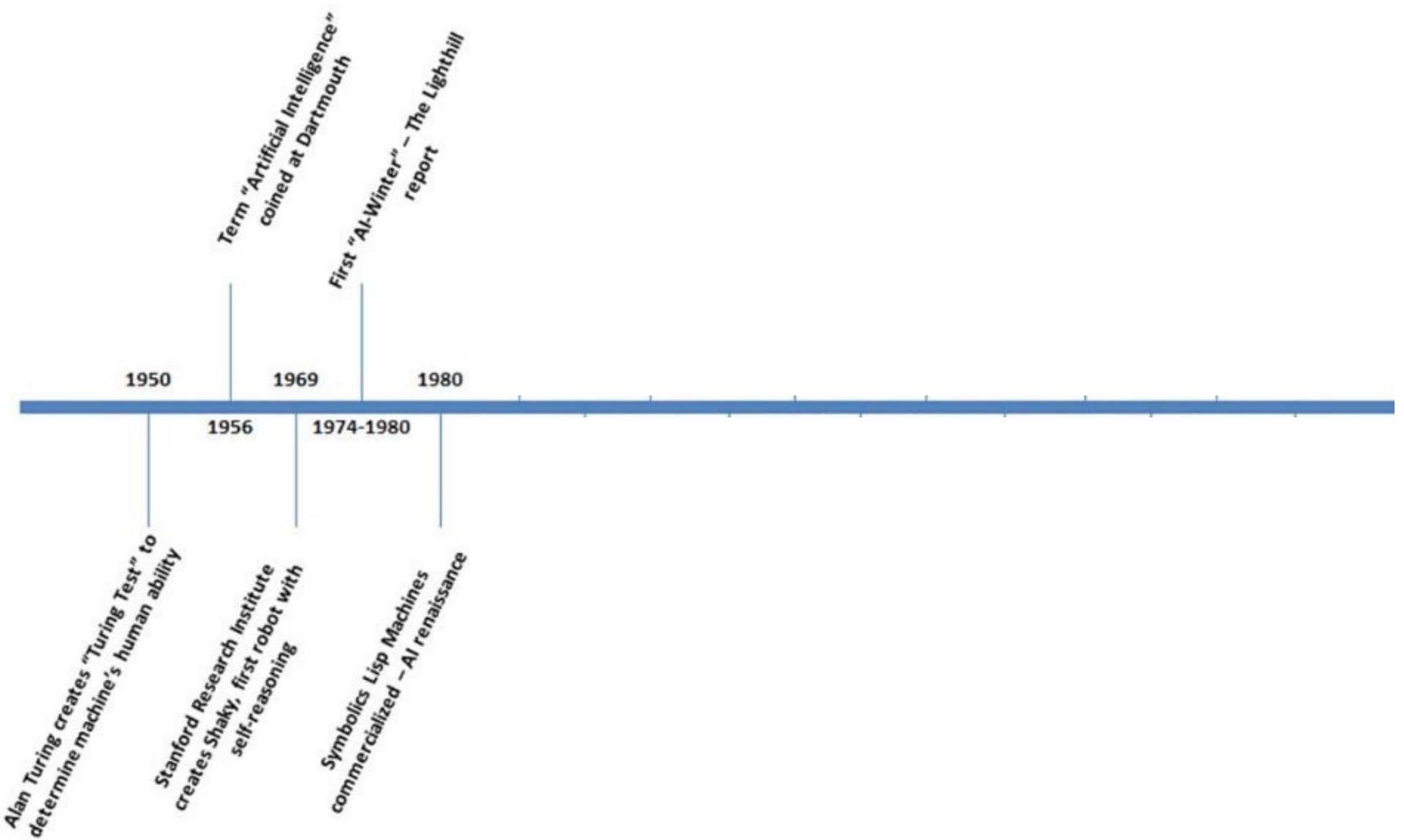

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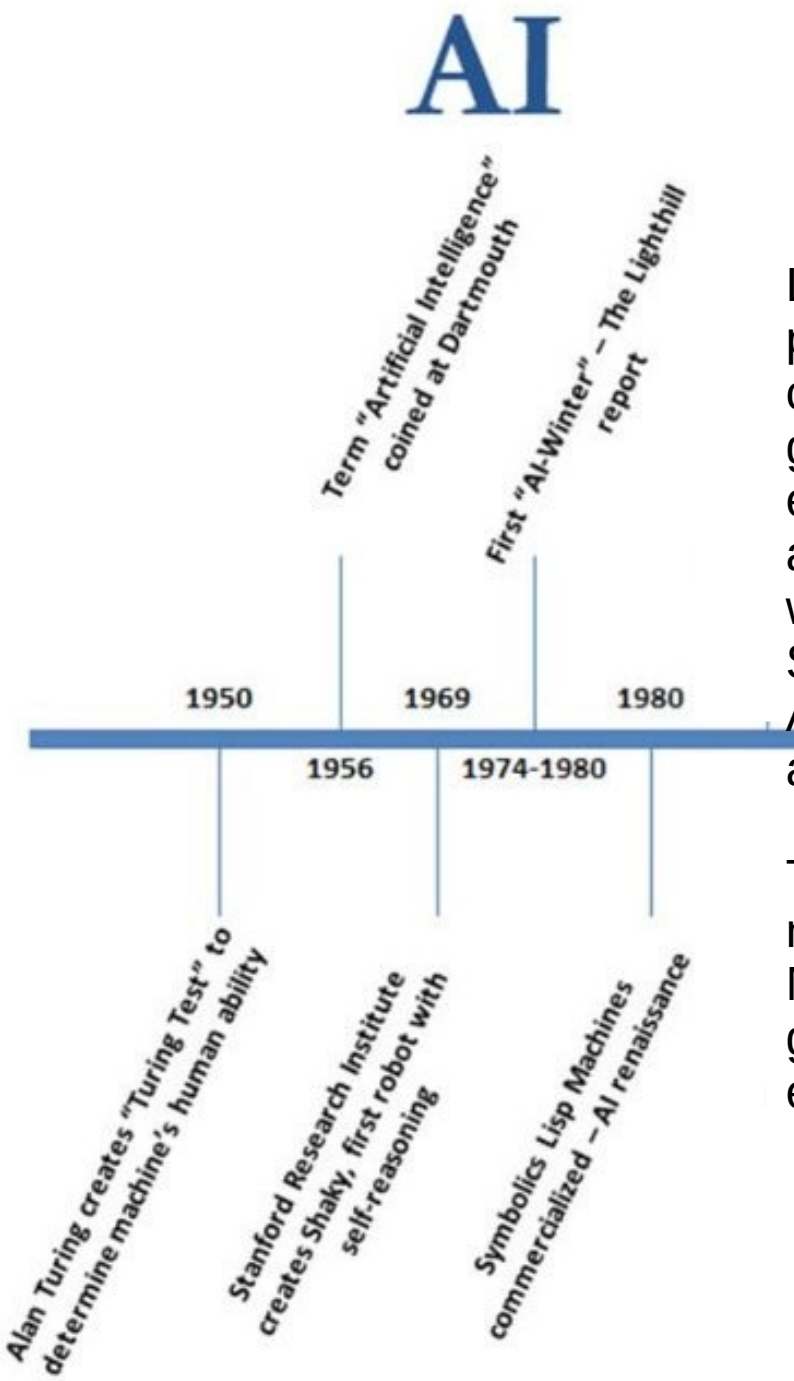


Lighthill Report

The Lighthill report is the name commonly used for the paper "Artificial Intelligence: A General Survey" by Professor Sir James Lighthill, published in Artificial Intelligence: a paper symposium in 1973.lighthill_report Published in 1973, it was compiled by Lighthill for the British Science Research Council, and evaluated academic research in the field of Artificial Intelligence. The report gave a very pessimistic prognosis for many core aspects of research in this field, stating that "in no part of the field have discoveries made so far produced the major impact that was then promised". It "formed the basis for the decision by the British government to end support for AI research in all but three universities"aiama—Edinburgh, Sussex and Essex. While the report was supportive of research into the simulation of neurophysiological and psychological processes, it was "highly critical of basic research in foundational areas such as robotics and language processing".ai_at_edinburgh The report stated that AI researchers had failed to address the issue of combinatorial explosion when solving problems within real world domains.

AI

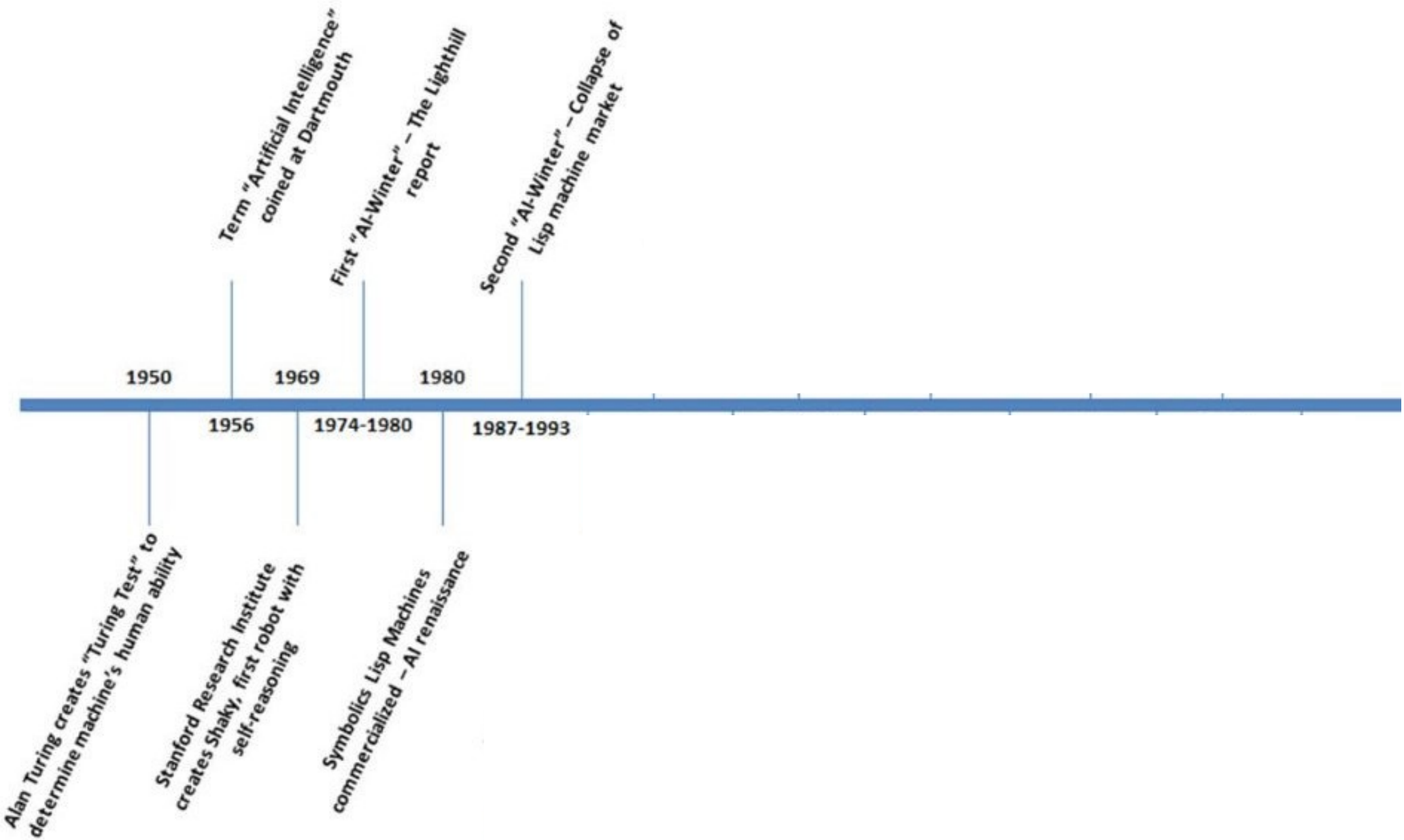

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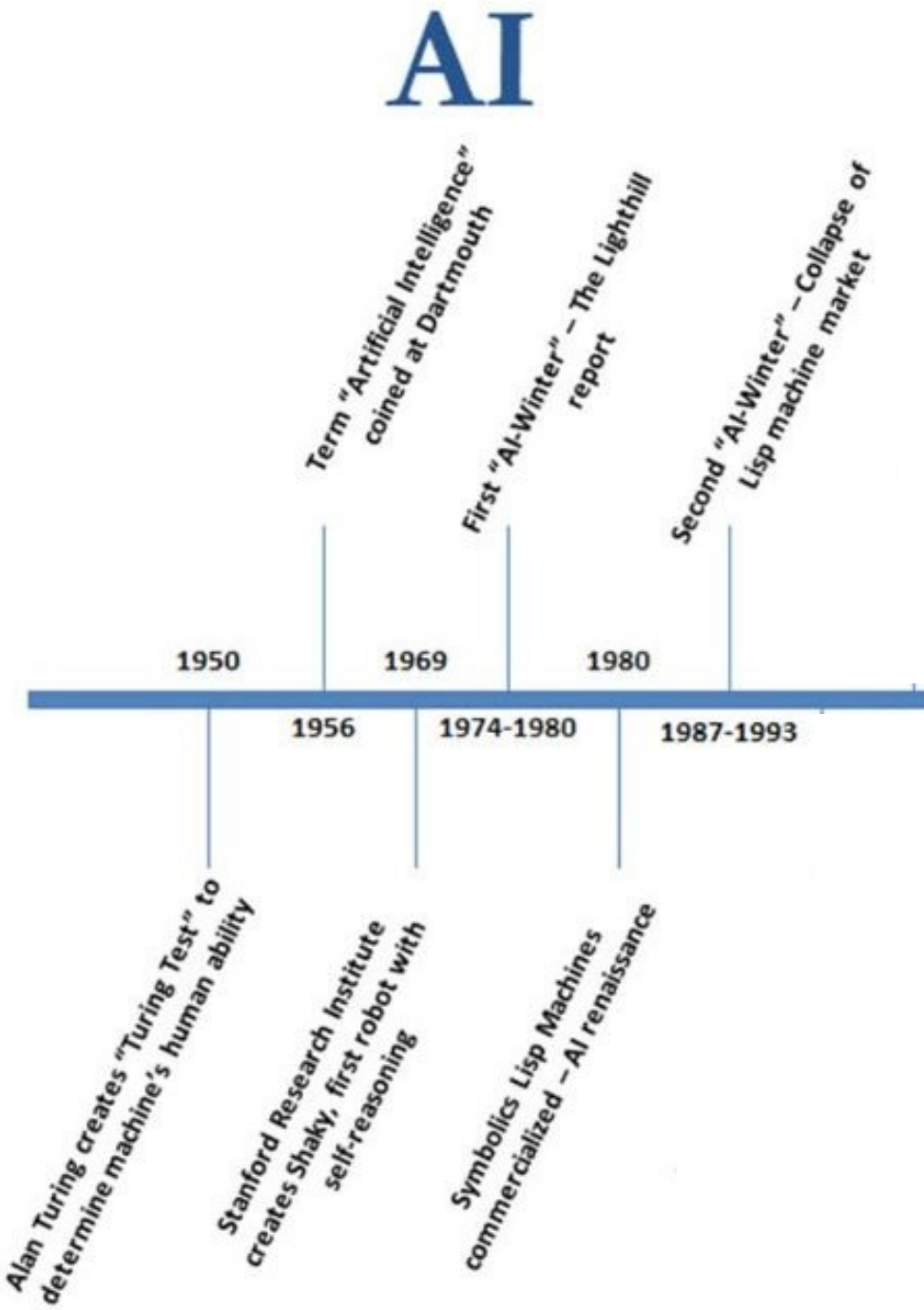


Lisp machines had hardware support for various primitive Lisp operations (data type testing, CDR coding) and also hardware support for incremental garbage collection. They ran large Lisp programs very efficiently. The Symbolics machine was competitive against many commercial super minicomputers, but was never adapted for conventional purposes. The Symbolics Lisp Machines were also sold to some non-AI markets like computer graphics, modeling, and animation.

The MIT-derived Lisp machines ran a Lisp dialect named Lisp Machine Lisp, descended from MIT's Maclisp. The operating systems were written from the ground up in Lisp, often using object-oriented extensions.

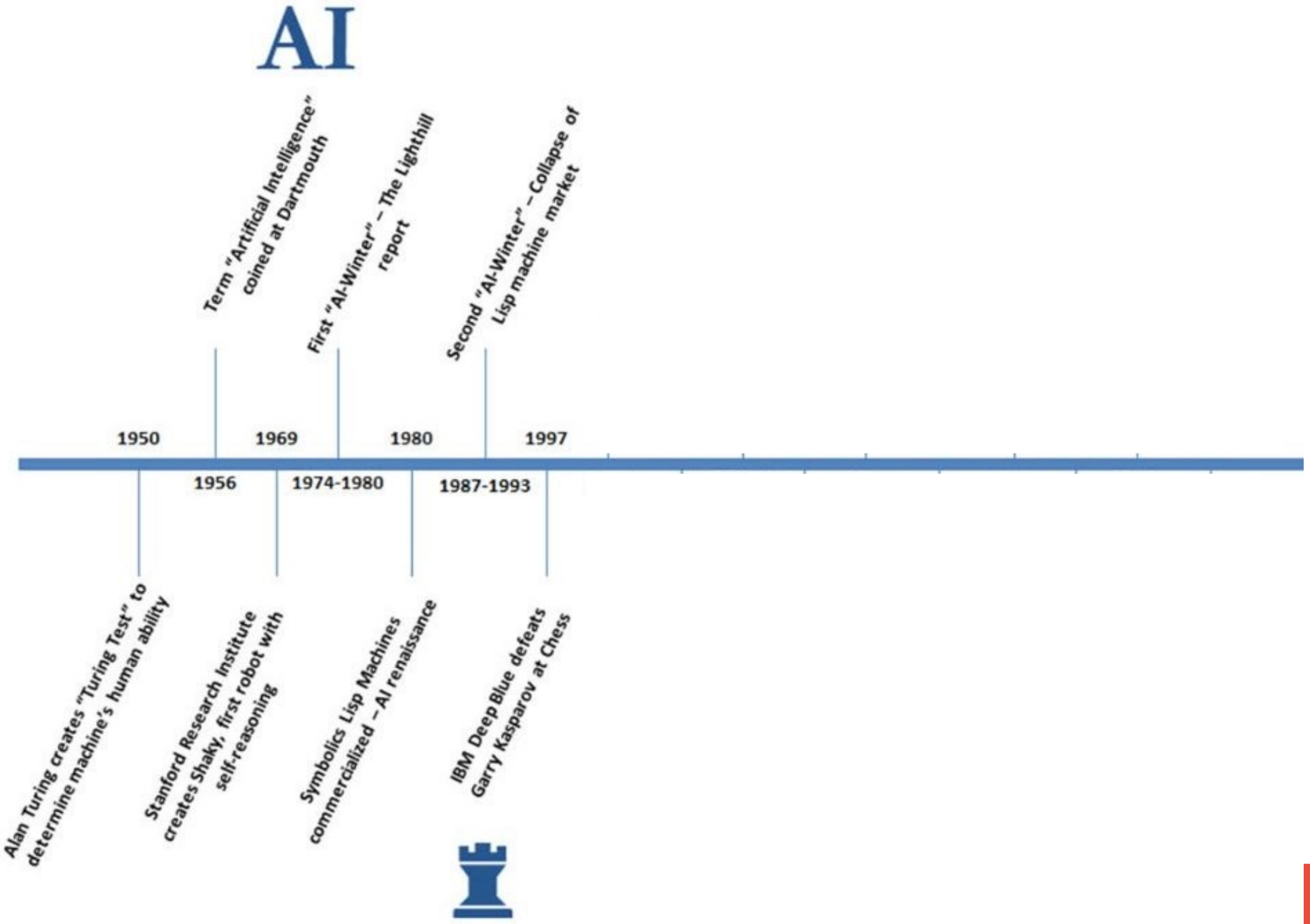
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With the early beginnings of the microcomputer revolution, which would sweep away the minicomputer and workstation makers, cheaper desktop PCs soon could run Lisp programs even faster than Lisp machines, with no use of special purpose hardware. Their high profit margin hardware business eliminated, most Lisp machine makers went out of business by the early 90s, leaving only software based firms like Lucid Inc. or hardware makers who switched to software and services to avoid the crash.

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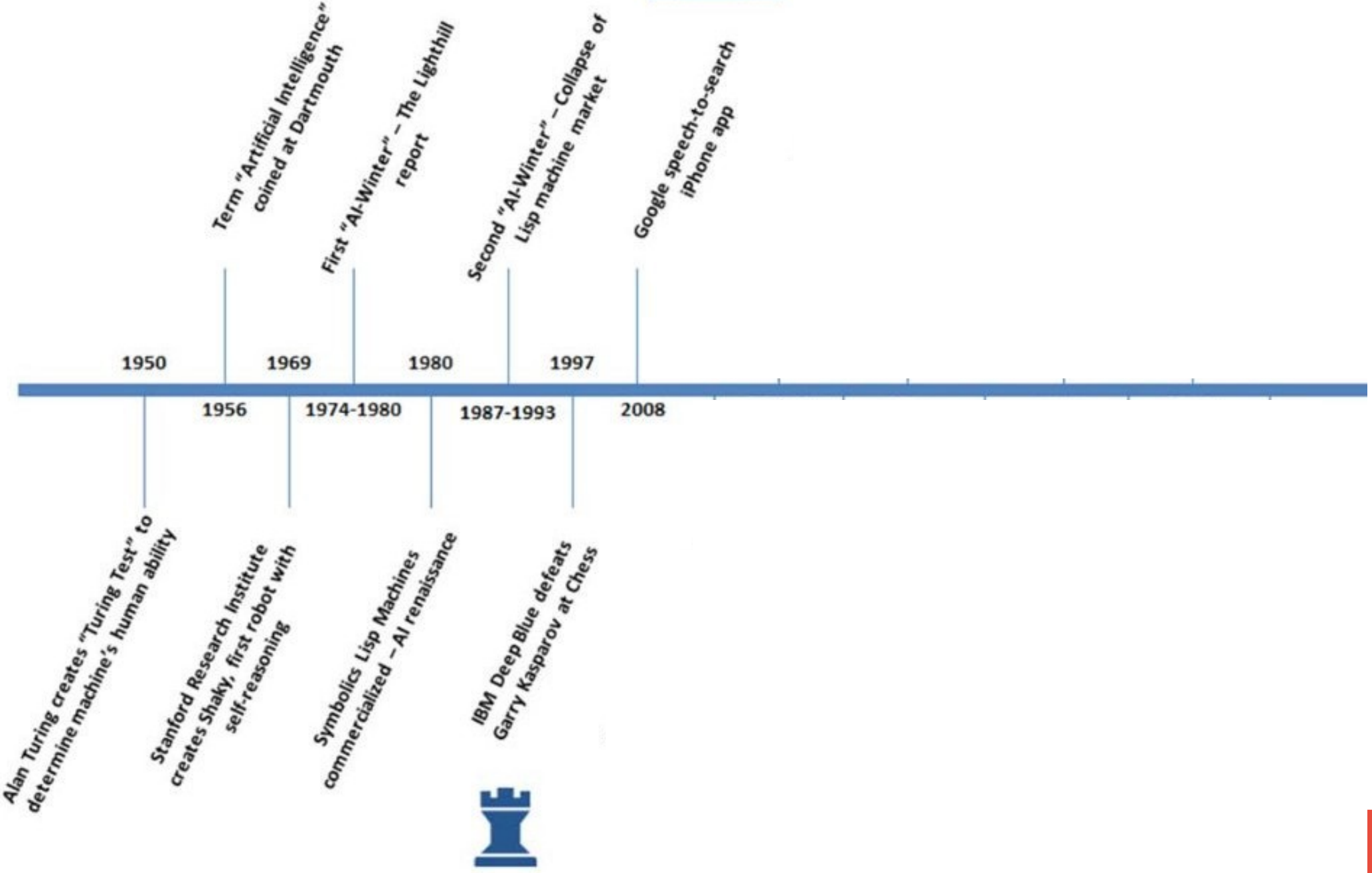


19

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AI



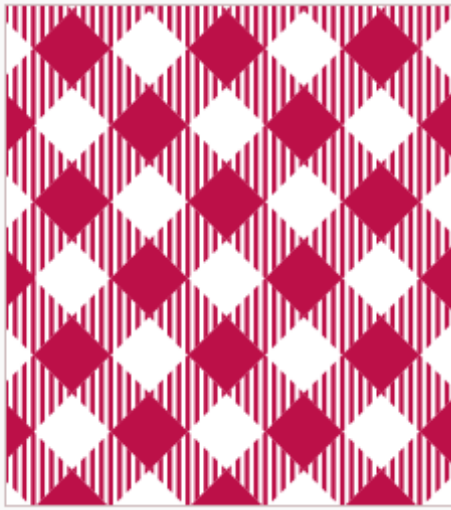

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Google to release speech-to-search iPhone app

*Posted Nov 14, 2008 by **Matt Burns** (@mjburnsy)*

Voice recognition has always seemed like the holy grail of technology and now Google is throwing their skills into the technology. Starting sometime today, iPhone users can download a free app that allows them to speak a destination or question and within seconds, receive an answer. The app works by recording your voice, uploading it to a Google Server via the data connection, crunching the data, and then returning the results. It doesn't deliever perfect results just yet, but as the app matures and learns everyones searching techniques, eventually it will creep closer to perfection.

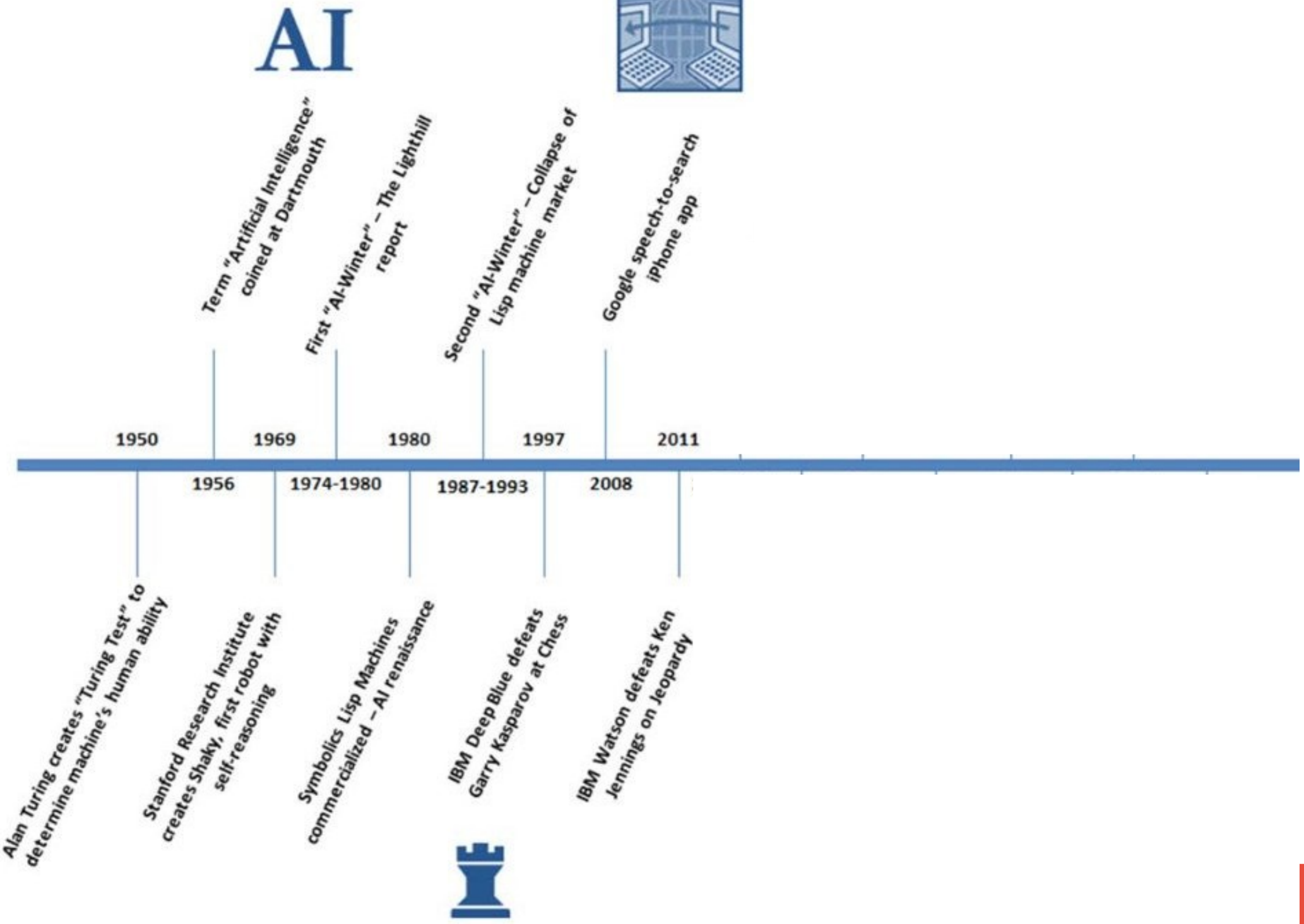
Kind of scary if you think about it. Google now reads your mail and knows what you sound like...



Crunchbase

AI




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AI


THINK
DENKE
$24,000
Who is Stoker?
((FOR ONE WELCOME OUR NEW COMPUTER OVERLORDS))
$1,000
$77,147
Who is Bram Stoker?
$ 17,973
$21,600
WHO IS BRAM STOKER?
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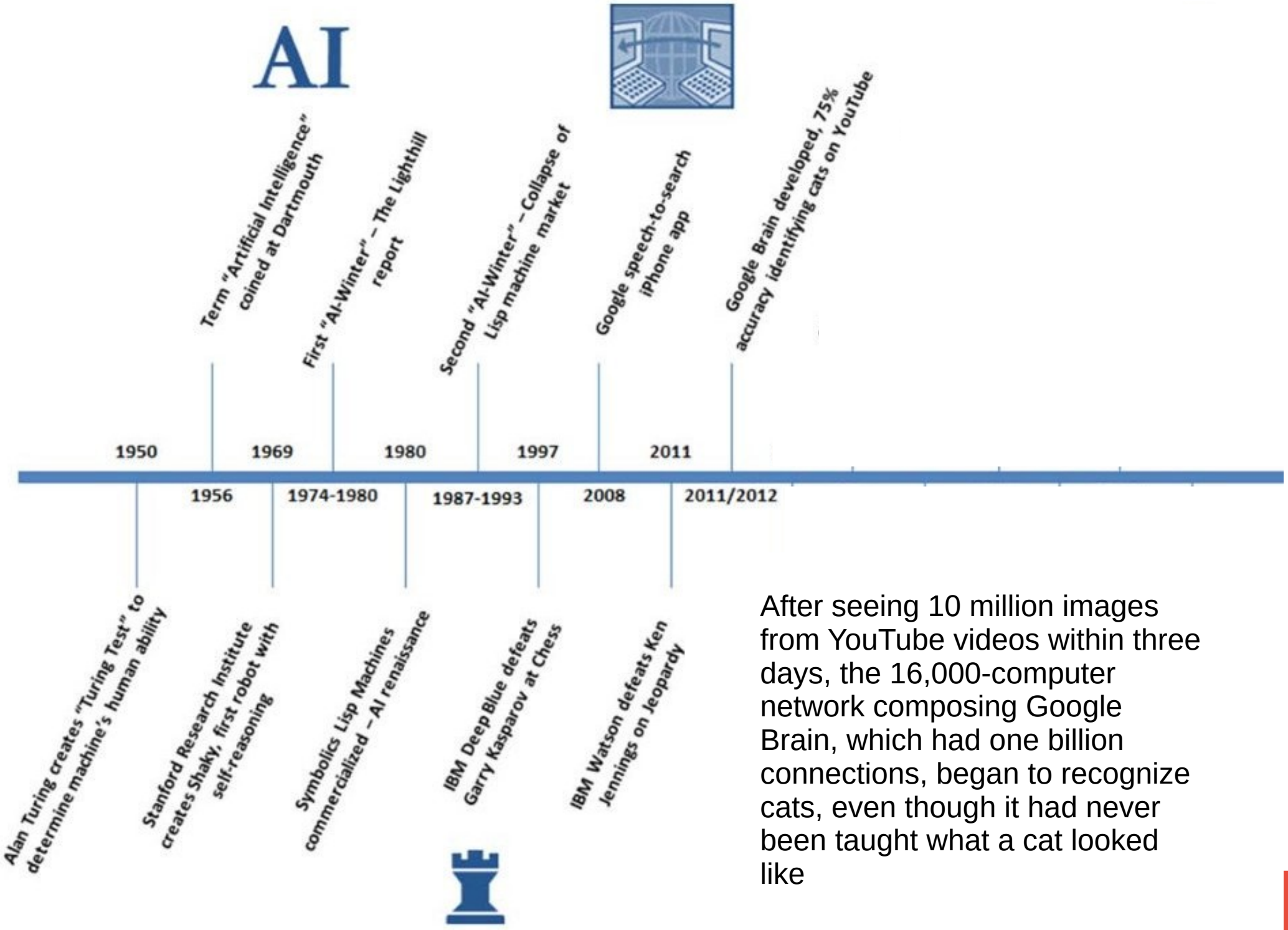
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2011/2012 — Google Brain developed, 75% accuracy identifying cats on YouTube





After seeing 10 million images from YouTube videos within three days, the 16,000-computer network composing Google Brain, which had one billion connections, began to recognize cats, even though it had never been taught what a cat looked like

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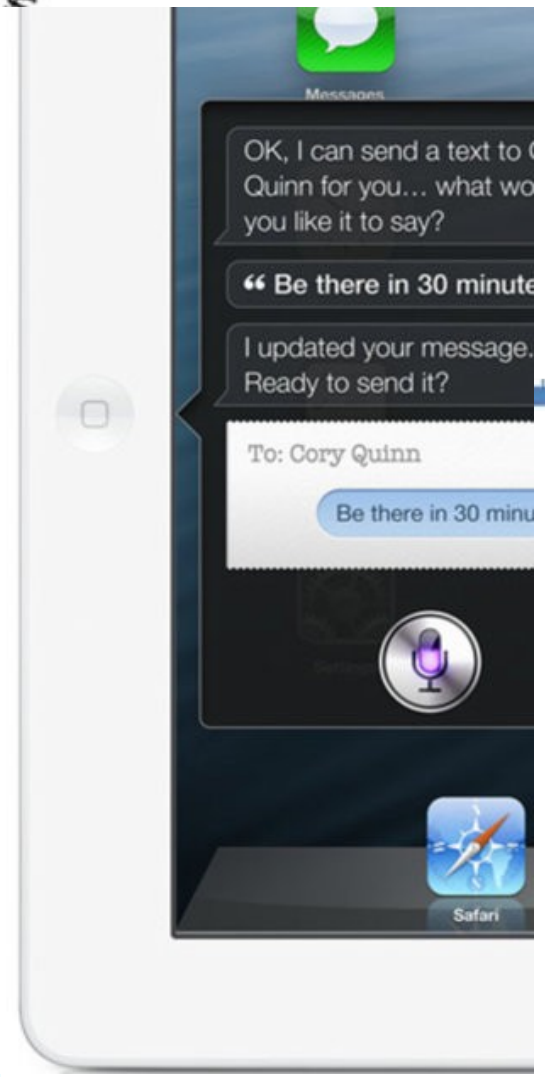


Siri. Beta

Your wish is its command.

Siri lets you use your voice to send messages, schedule meetings, place phone calls, and more.* Ask Siri to do things just by talking the way you talk. Siri is so easy to use and does so much, you'll keep finding more and more ways to use it.





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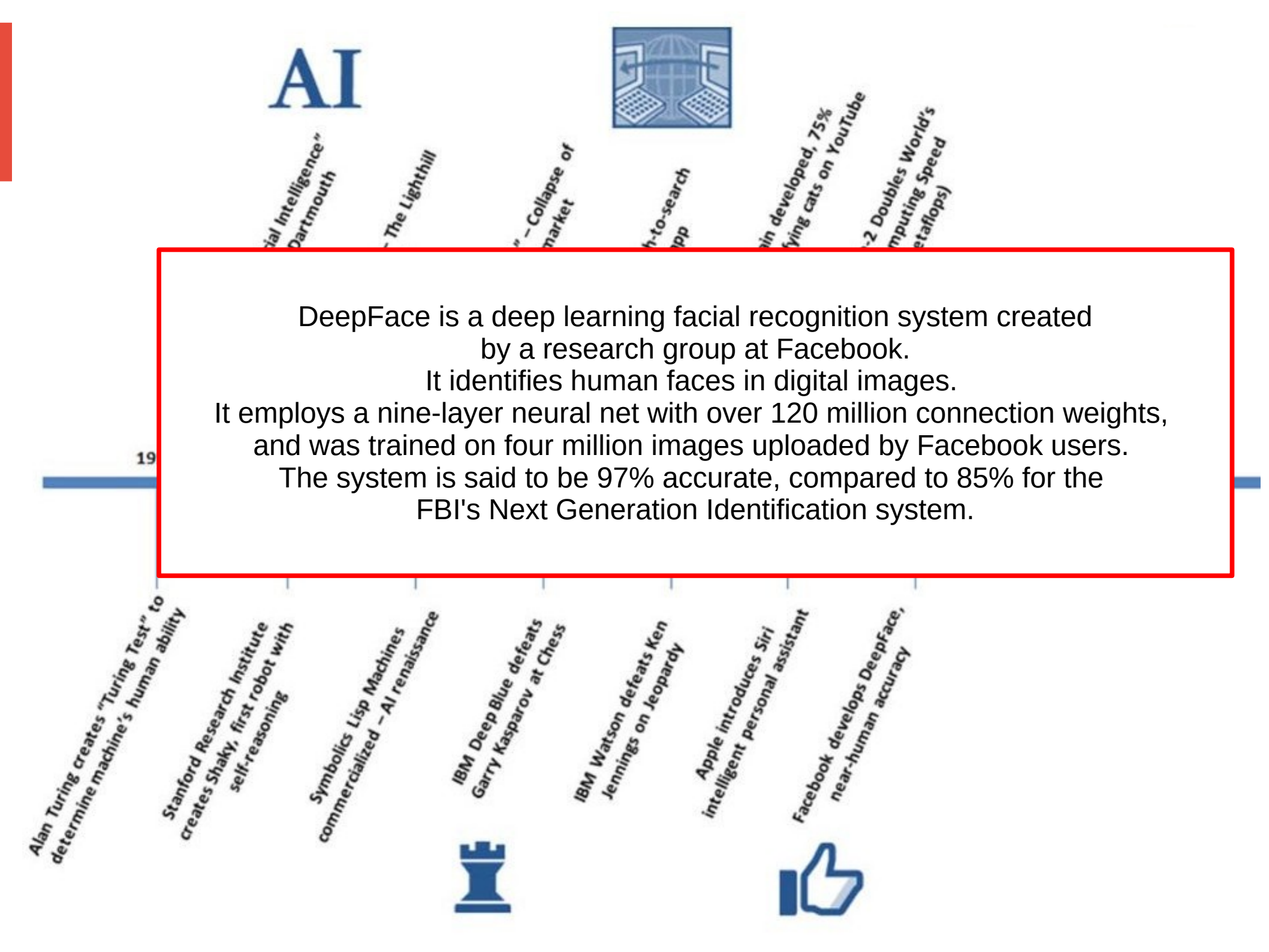
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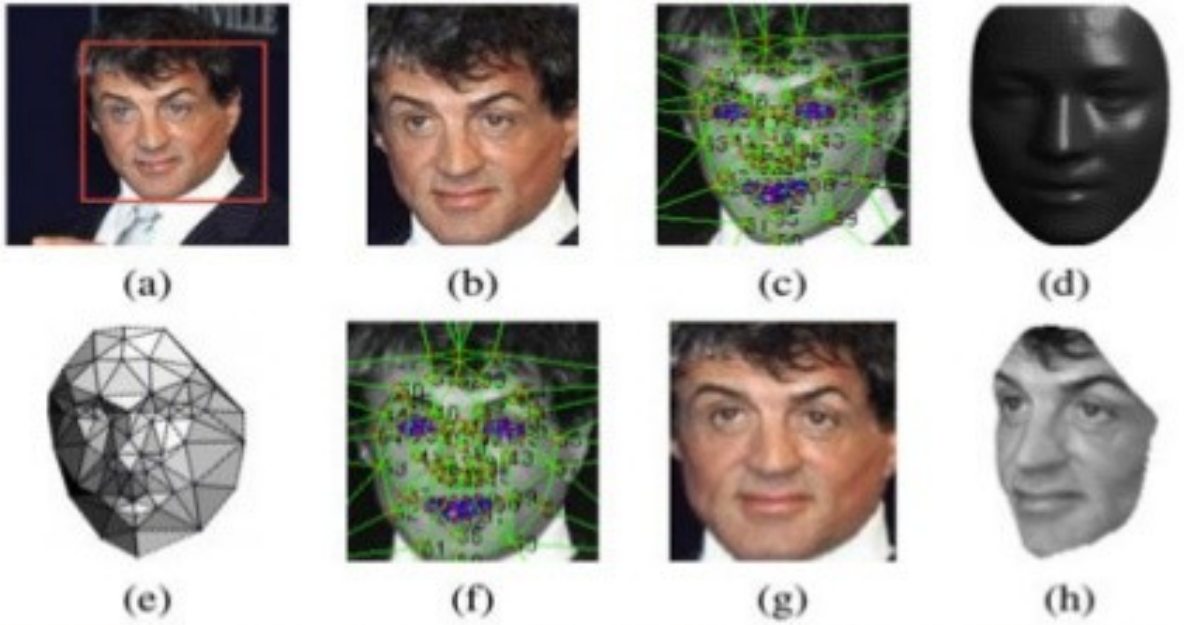


2014 — Facebook develops DeepFace, near-human accuracy


AI
DeepFace is a deep learning facial recognition system created by a research group at Facebook.
It identifies human faces in digital images.
It employs a nine-layer neural net with over 120 million connection weights, and was trained on four million images uploaded by Facebook users.
The system is said to be 97% accurate, compared to 85% for the FBI's Next Generation Identification system.
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Breakthrough Two

- DeepFace by Facebook AI Research



DeepFace uses a 3-D model to rotate faces, virtually, so that they face the camera. Image (a) shows the original image, and (g) shows the final, corrected version.

Alan Turing creates "
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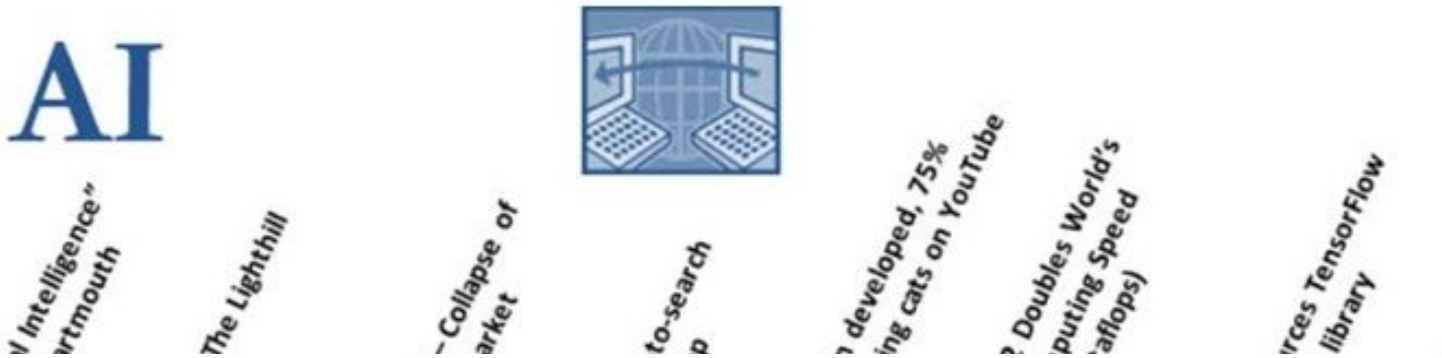
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2015 – Google open-sources TensorFlow software library



About TensorFlow

TensorFlow™ is an open source software library for numerical computation using data flow graphs. Nodes in the graph represent mathematical operations, while the graph edges represent the multidimensional data arrays (tensors) communicated between them. The flexible architecture allows you to deploy computation to one or more CPUs or GPUs in a desktop, server, or mobile device with a single API. TensorFlow was originally developed by researchers and engineers working on the Google Brain Team within Google's Machine Intelligence research organization for the purposes of conducting machine learning and deep neural networks research, but the system is general enough to be applicable in a wide variety of other domains as well.







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1987-1993 — Second "AI-Winter" – Collapse of Lisp machine market



1997 — IBM Deep Blue defeats Garry Kasparov at Chess

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2016 — Google open-sources TensorFlow software library

2016 — China's Sunway TaihuLight Triples World's Top Supercomputing Speed (93 petaflops)



2016 — Uber pilots self-driving car program in Pittsburgh, PA

AI



1950

2016

Alan Turing creates "Turing Test" to determine machine's human abil

Pittsburgh, PA

AI





1950 — Alan Turing creates "Turing Test" to determine machine's human ability

1956 — Term "Artificial Intelligence" coined at Dartmouth

1969 — Stanford Research Institute creates Shaky, first robot with self-reasoning

1974-1980 — First "AI-Winter" – The Lighthill report

1980 — Symbolics Lisp Machines commercialized – AI renaissance

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2016 — Uber pilots self-driving car program in Pittsburgh, PA

Some more recent achievements

https://www.youtube.com/watch?v=PFMRDm_H9Sg



AlphaStar won its historic series against professional [StarCraft II players on January 24, 2019], after previously playing the pre-recorded matches in [December 2018].

Some more recent achievements

https://www.nature.com/articles/s41591-019-0447-x

Letter | Published: 20 May 2019

End-to-end lung cancer screening with three-dimensional deep learning on low-dose chest computed tomography

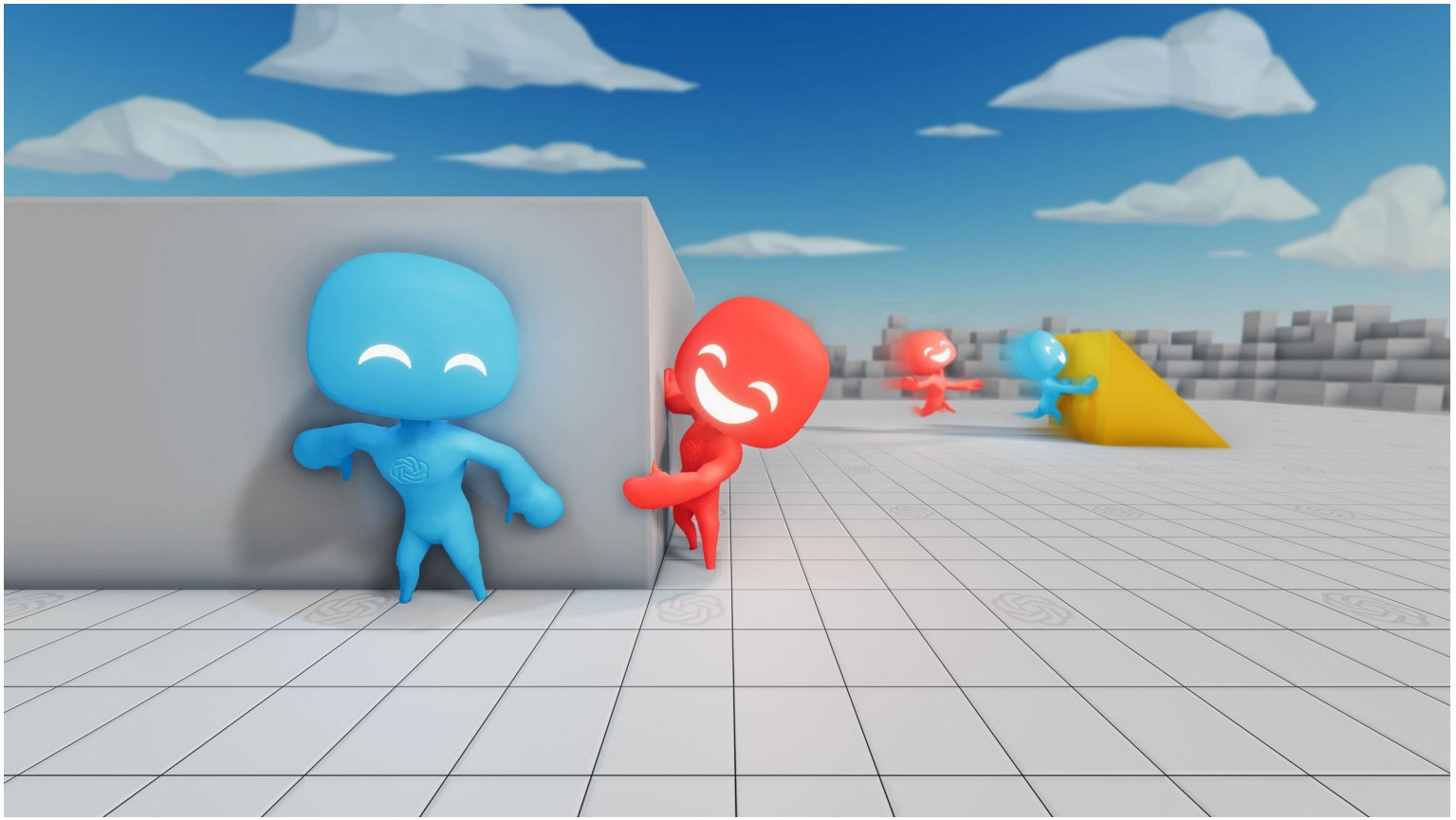
Diego Ardila, Atilla P. Kiraly, Sujeeth Bharadwaj, Bokyung Choi, Joshua J. Reicher, Lily Peng, Daniel Tse ✉, Mozziyar Etemadi, Wenxing Ye, Greg Corrado, David P. Naidich & Shravya Shetty

Nature Medicine **25**, 954–961(2019) | Cite this article

38k Accesses | **150** Citations | **1584** Altmetric | Metrics

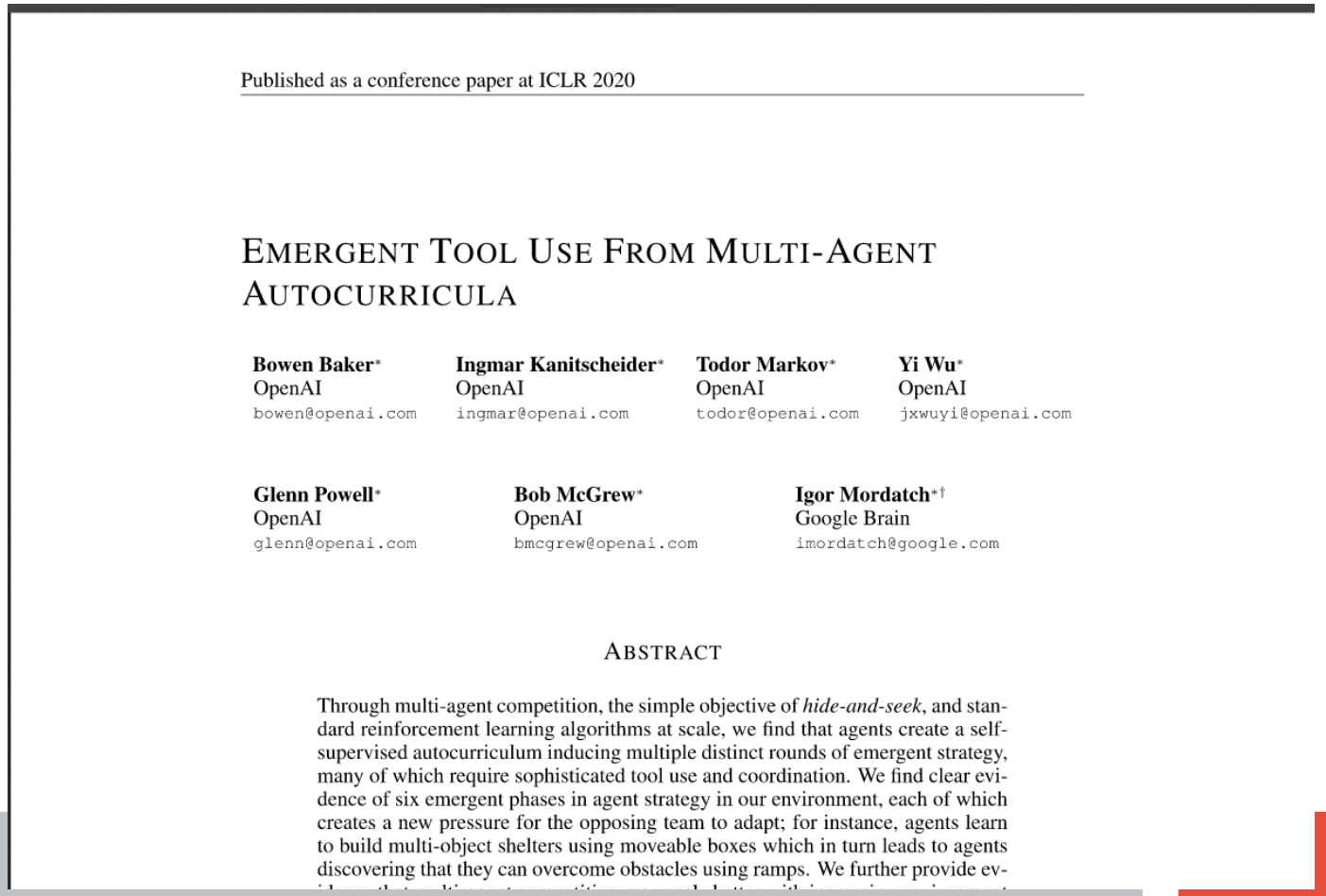
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https://www.youtube.com/watch?v=kopoLzvh5jY



Some more recent achievements

https://www.youtube.com/watch?v=kopoLzvh5jY



Published as a conference paper at ICLR 2020

EMERGENT TOOL USE FROM MULTI-AGENT AUTOCURRICULA

Bowen Baker* OpenAI bowen@openai.com

Ingmar Kanitscheider* OpenAI ingmar@openai.com

Todor Markov* OpenAI todor@openai.com

Yi Wu* OpenAI jxwuyi@openai.com

Glenn Powell* OpenAI glenn@openai.com

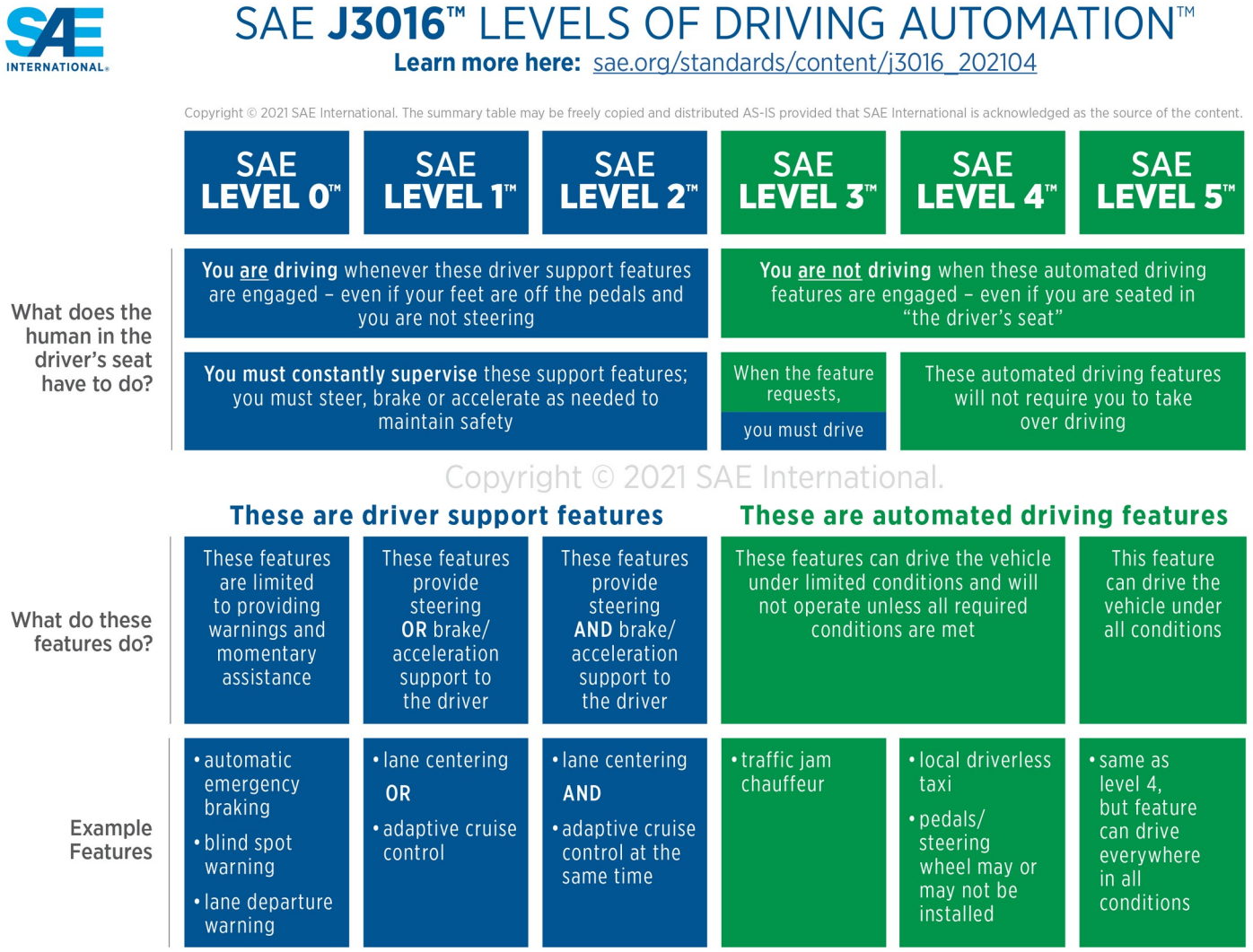
Bob McGrew* OpenAI bmcgrew@openai.com

Igor Mordatch*† Google Brain imordatch@google.com

ABSTRACT

Through multi-agent competition, the simple objective of *hide-and-seek*, and standard reinforcement learning algorithms at scale, we find that agents create a self-supervised autocurriculum inducing multiple distinct rounds of emergent strategy, many of which require sophisticated tool use and coordination. We find clear evidence of six emergent phases in agent strategy in our environment, each of which creates a new pressure for the opposing team to adapt; for instance, agents learn to build multi-object shelters using moveable boxes which in turn leads to agents discovering that they can overcome obstacles using ramps. We further provide ev-

Some more recent achievements



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	SAE LEVEL 0™	SAE LEVEL 1™	SAE LEVEL 2™	SAE LEVEL 3™	SAE LEVEL 4™	SAE LEVEL 5™
What does the human in the driver's seat have to do?	**You are driving** whenever these driver support features are engaged – even if your feet are off the pedals and you are not steering			**You are not driving** when these automated driving features are engaged – even if you are seated in "the driver's seat"		
	You must constantly supervise these support features; you must steer, brake or accelerate as needed to maintain safety			When the feature requests, you must drive	These automated driving features will not require you to take over driving	
		These are driver support features		**These are automated driving features**		
What do these features do?	These features are limited to providing warnings and momentary assistance	These features provide steering **OR** brake/ acceleration support to the driver	These features provide steering **AND** brake/ acceleration support to the driver	These features can drive the vehicle under limited conditions and will not operate unless all required conditions are met		This feature can drive the vehicle under all conditions
Example Features	• automatic emergency braking • blind spot warning • lane departure warning	• lane centering **OR** • adaptive cruise control	• lane centering **AND** • adaptive cruise control at the same time	• traffic jam chauffeur	• local driverless taxi • pedals/ steering wheel may or may not be installed	• same as level 4, but feature can drive everywhere in all conditions

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Some more recent achievements

What You Need to Know About Level 3 Autonomy

You can now buy a Level 3 car in the U.S. — if you live in one of two states and can afford a new Mercedes-Benz.

David Gluckman | Jun 21, 2024 | 6 min read



Some more recent achievements

Mercedes-Benz

QuickTakes:

Level 3 is a step toward full autonomy
Automated driving requires lots of sensors
Mercedes-Benz is the only company offering Level 3 vehicles
Wide adoption of Level 3 will require regulators to catch up

Automakers have been promising fully autonomous cars for years, but only now are the technology and necessary regulations catching up. That said, not all autonomous vehicles are the same, and the first cars that let you pay attention to something other than driving — vehicles with so-called **Level 3** driving automation — have limited application. They work on only a small selection of roads, and they're sold only in certain states.



Some more recent achievements

which are the three most impressive achievements in the Artificial Intelligence research field, in the last three years (short answer)

1. **Generative AI Advancements**: Models like GPT-4 and DALL·E have revolutionized AI with their ability to generate coherent text, images, and creative content, pushing the boundaries of creativity and human-like understanding.
2. **AI in Protein Folding**: DeepMind's AlphaFold achieved groundbreaking success by accurately predicting protein structures, solving a 50-year-old problem in biology, with huge implications for drug discovery and healthcare.
3. **AI Alignment and Safety Research**: Progress in AI interpretability, alignment, and safety, including work from OpenAI and Anthropic, has made strides in controlling the behavior of advanced AI systems, addressing ethical and safety concerns.

Some more recent achievements

https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2443096/

The protein folding problem is the question of how a protein's amino acid sequence dictates its three-dimensional atomic structure. The notion of a folding "problem" first emerged around 1960, with the appearance of the first atomic-resolution protein structures.

Some more recent achievements

Signal Transduction and Targeted Therapy

www.nature.com/sigtrans



REVIEW ARTICLE OPEN

AlphaFold2 and its applications in the fields of biology and medicine

Zhenyu Yang[1], Xiaoxi Zeng[1]✉, Yi Zhao[1,2]✉ and Runsheng Chen[1,3,4]✉

AlphaFold2 (AF2) is an artificial intelligence (AI) system developed by DeepMind that can predict three-dimensional (3D) structures of proteins from amino acid sequences with atomic-level accuracy. Protein structure prediction is one of the most challenging problems in computational biology and chemistry, and has puzzled scientists for 50 years. The advent of AF2 presents an unprecedented progress in protein structure prediction and has attracted much attention. Subsequent release of structures of more than 200 million proteins predicted by AF2 further aroused great enthusiasm in the science community, especially in the fields of biology and medicine. AF2 is thought to have a significant impact on structural biology and research areas that need protein structure information, such as drug discovery, protein design, prediction of protein function, et al. Though the time is not long since AF2 was developed, there are already quite a few application studies of AF2 in the fields of biology and medicine, with many of them having preliminarily proved the potential of AF2. To better understand AF2 and promote its applications, we will in this article summarize the principle and system architecture of AF2 as well as the recipe of its success, and particularly focus on reviewing its applications in the fields of biology and medicine. Limitations of current AF2 prediction will also be discussed.

Signal Transduction and Targeted Therapy (2023)8:115 ; https://doi.org/10.1038/s41392-023-01381-z

Some more recent achievements

PNAS PERSPECTIVE OPEN ACCESS

AlphaFold two years on: Validation and impact

Oleg Kovalevskiy[1], Juan Mateos-Garcia, and Kathryn Tunyasuvunakool

Edited by Ulyana Shimanovich, Weizmann Institute of Science Department of Materials and Interfaces, Rehovot, Israel; received October 18, 2023; accepted March 8, 2024 by Editorial Board Member Ulrich Hartl

August 12, 2024 | 121 (34) e2315002121

Two years on from the initial release of AlphaFold, we have seen its widespread adoption as a structure prediction tool. Here, we discuss some of the latest work based on AlphaFold, with a particular focus on its use within the structural biology community. This encompasses use cases like speeding up structure determination itself, enabling new computational studies, and building new tools and workflows. We also look at the ongoing validation of AlphaFold, as its predictions continue to be compared against large numbers of experimental structures to further delineate the model's capabilities and limitations.

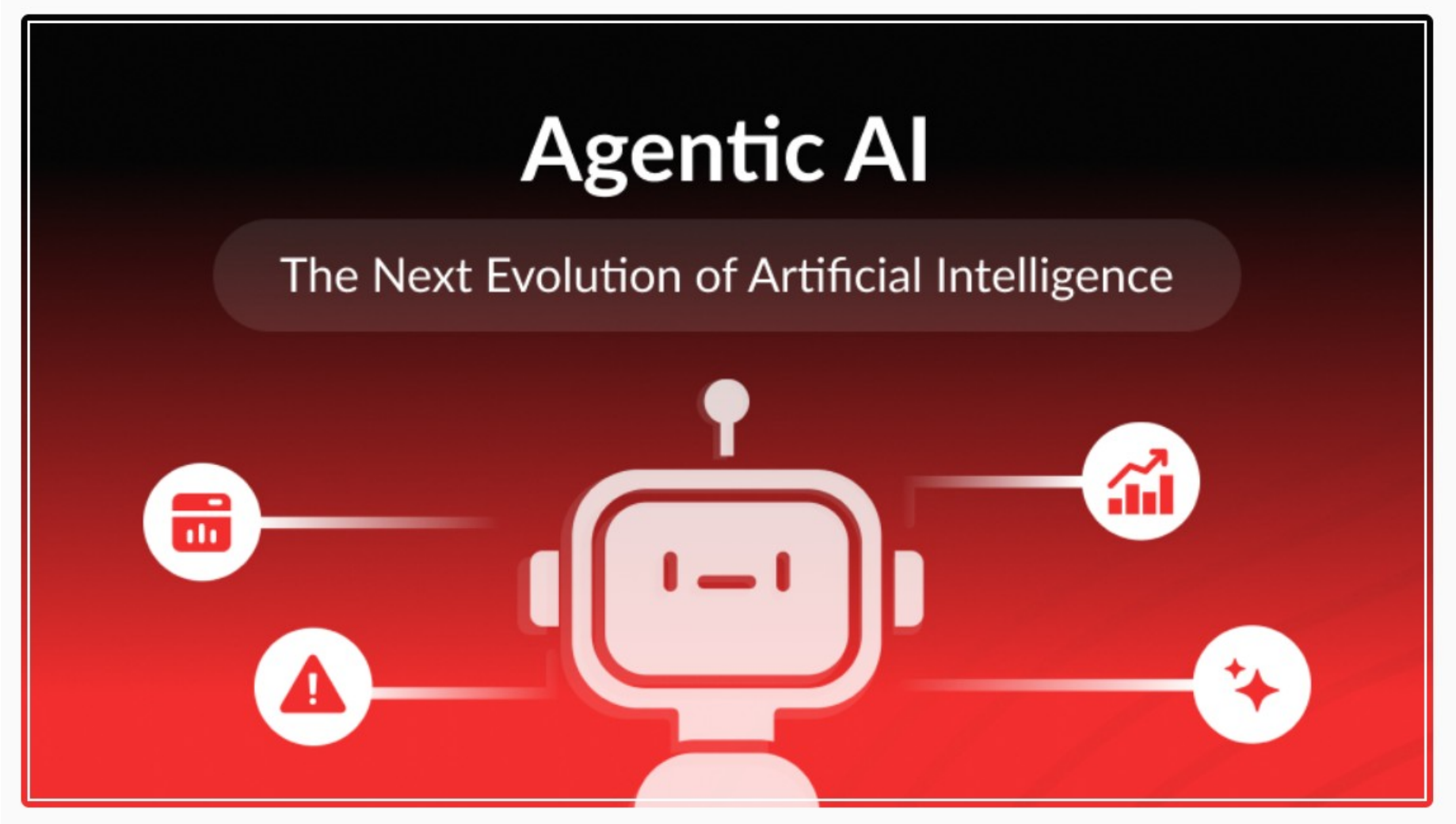
AlphaFold | structural biology | protein structure prediction

Google DeepMind, London N1C 4DN, United Kingdom

[1] To whom correspondence may be addressed. Email: okovalevskiy@deepmind.com.

Competing interests statement: All authors at the time of writing are employees of Google DeepMind, which has filed non-provisional patent applications: 16/701,070; PCT/EP2020/084238; PCT/EP2021/072552; EP21766396.2; US18/025,689; CN202180067160.6; PCT/EP2021/082684; EP21816053.9; US18/026,376;

And now… Agentic AI



But

https://harvardlawreview.org/2017/03/state-v-loomis/



HLR ABOUT BLOG FELLOWSHIP FORUM ISSUES ORDER SUBMIT SCOTUS STATISTICS SE

HARVARD LAW REVIEW

CRIMINAL LAW

State v. Loomis

Wisconsin Supreme Court Requires Warning Before Use of Algorithmic Risk Assessments in Sentencing.

Recent Case : 881 N.W.2d 749 (Wis. 2016)

MAR 10, 2017
130 Harv. L. Rev. 1530

% # p **PDF**

Judges rightly view sentencing as a weighty responsibility.[1] They must consider not only the appropriate punishment for the offense but also the risk the offender poses, predicting the probability of the offender's recidivism.[2] A potential solution to this judicial anxiety has arrived thanks to the bipartisan interest in criminal

1. *See, e.g.,* Williams v. New York, 337 U.S.

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(Image credit: TechNode/Cassidy McDonald)

Shanghai Court Adopts New AI Assistant – Sixth Tone

What happened: A Shanghai court has adopted an artificial intelligence-enabled assistant to help improve courtroom efficiency and accuracy. The city’s No. 2 Immediate People’s Court is the first in the country to utilize the system, dubbed System 206, developed by Chinese tech firm iFlytech and the country’s judicial and public

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OpenAI launched ChatGPT last November

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OpenAI's Altman and Anthropic's Amodei address UN security council

Heads of two of the world's largest artificial intelligence companies give separate briefings on AI safety



Dario Amodei addresses the United Nations security council in New York City on Wednesday. Photograph: Brendan McDermid/Reuters

Sam Altman of OpenAI and Dario Amodei of Anthropic, heads of two of the world's largest artificial intelligence companies, addressed the United Nations security council on Wednesday in separate briefings on AI safety.

"We have a choice in front of us," Altman told the council. "AI can either be more like a new renaissance of creativity and discovery, or more like a new industrial revolution of upheaval and disarray."

Altman and Amodei spoke as their companies face increasing scrutiny over the safety of their products. Public alarm over AI has grown in response to warnings from Anthropic researchers earlier this month that the technology could destroy humanity within the decade. Leaders of AI companies,

But

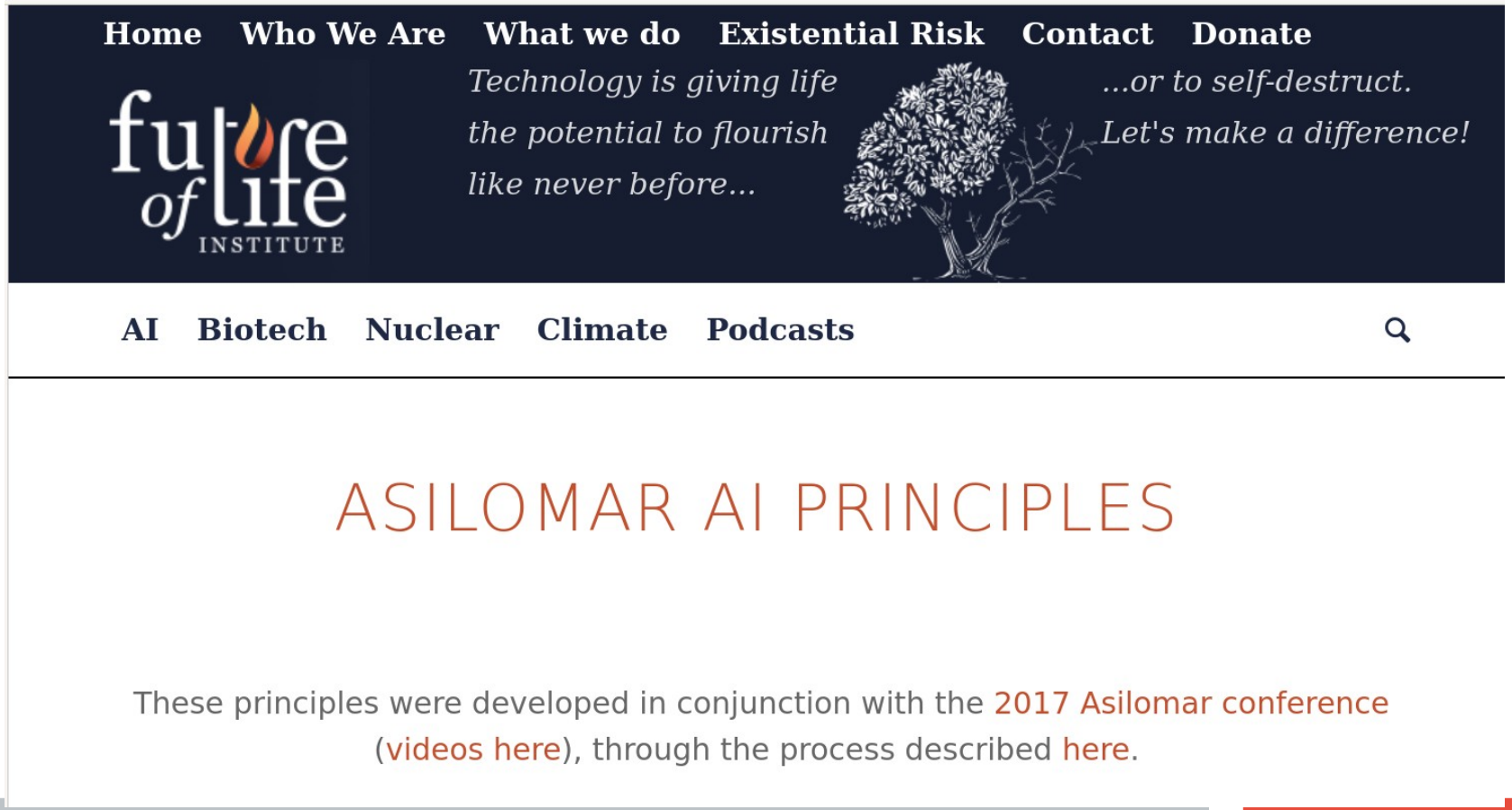
https://www.bbc.com/news/articles/cz7dl7w8y7po

https://openai.com/index/hugging-face-incident-and-the-road-ahead/

https://www.reuters.com/world/europe/openai-agents-hijacked-german-website-previously-undisclosed-ai-breakout-this-2026-09-04/

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https://futureoflife.org/ai-principles/



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https://ec.europa.eu/digital-single-market/en/news/ethics-guidelines-trustworthy-ai



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Shaping Europe’s digital future

REPORT / STUDY | 8 April 2019

Ethics guidelines for trustworthy AI

On 8 April 2019, the High-Level Expert Group on AI presented Ethics Guidelines for Trustworthy Artificial Intelligence. This followed the publication of the

About Artificial intelligence

Blog posts

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https://www.nytimes.com/2026/09/12/technology/anthropic-dario-amodei-ai-slowdown.html

Top A.I. Leaders Call for Slowing Down A.I. Development

In an essay, the head of Anthropic laid out the rapidly advancing capabilities of artificial intelligence and said there needed to be greater safety controls across the industry.

Listen · 7:38 min

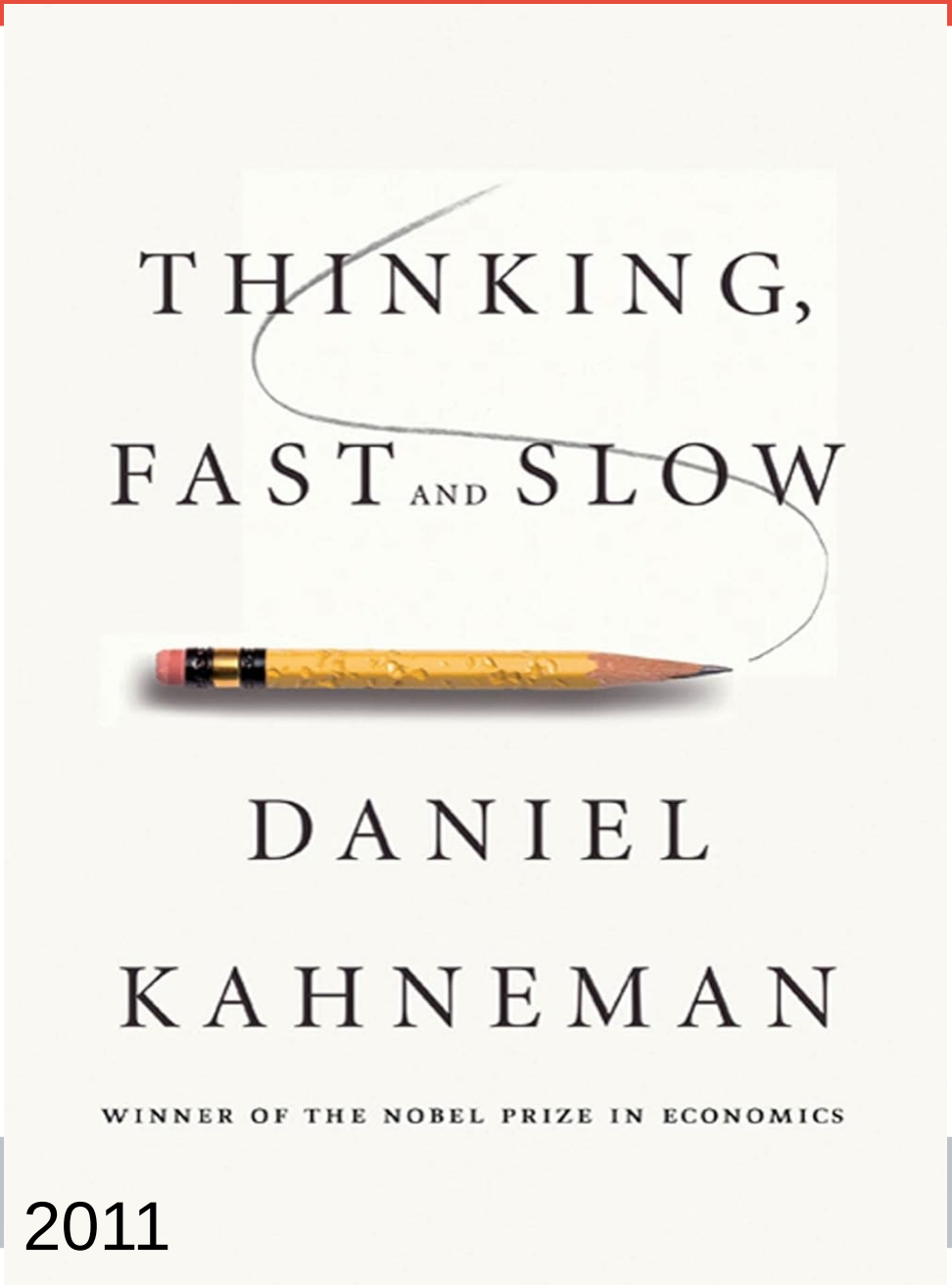
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Dario Amodei, Anthropic's chief executive, in February. "Over the last few months, I have become convinced that fully addressing the risks requires even more prudence," he wrote in an essay. Ludovic Marin/Agence France-Presse — Getty Images

Back to the first slides



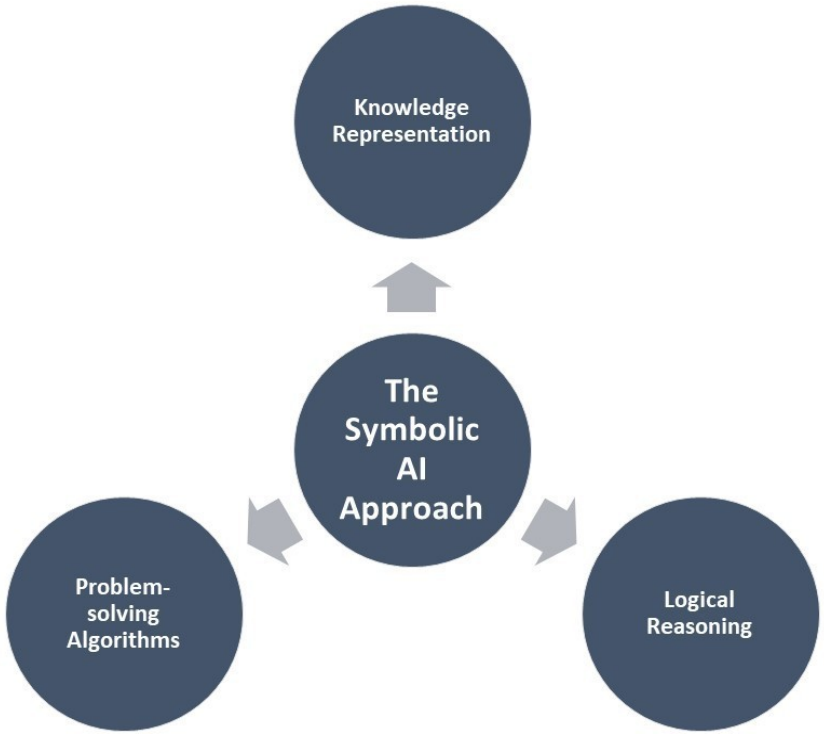
2011

Sub-Symbolic AI



https://www.linkedin.com/pulse/bridging-gap-between-symbolic-subsymbolic-ai-prof-ahmed-banafa-ujkjc/

Symbolic (“Good Old Fashioned”) AI



https://www.linkedin.com/pulse/bridging-gap-between-symbolic-subsymbolic-ai-prof-ahmed-banafa-ujkjc/

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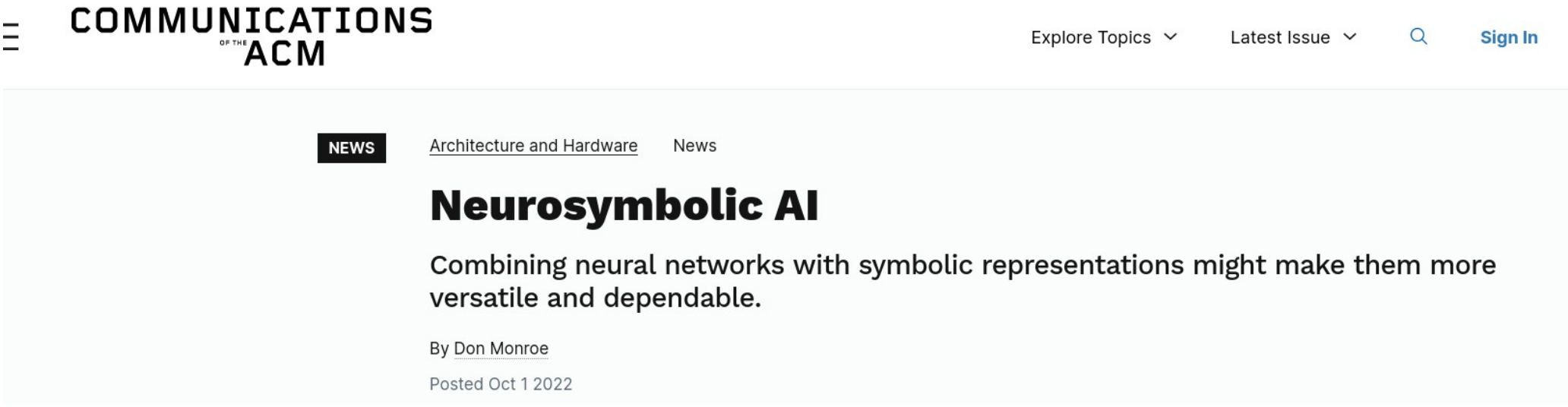
Grady Booch, Francesco Fabiano, Lior Horesh, Kiran Kate, Jonathan Lenchner, Nick Linck, Andrea Loreggia, Keerthiram Murugesan, Nicholas Mattei, Francesca Rossi, Biplav Srivastava:
Thinking Fast and Slow in AI. AAAI 2021: 15042-15046

System 1 and System 2 in AI

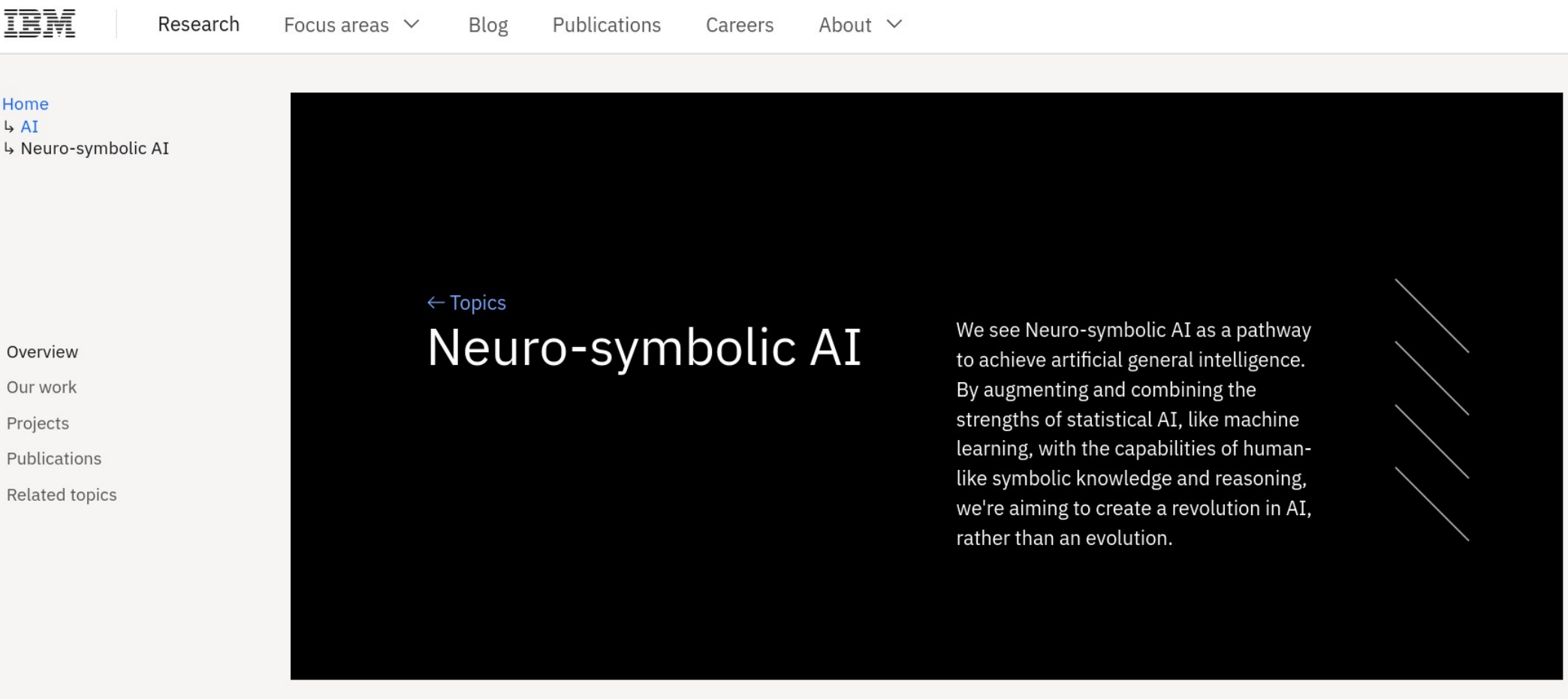
It is possible to draw a very loose parallel between the capabilities included in system 1 and system 2, and the two main lines of work in AI: machine learning (ML) and symbolic logic reasoning, or data-driven vs knowledge-driven AI. System 1 seems to have similar traits as ML, with its ability to build (possibly imprecise and biased) models from sensory data. For example, perception activities, such as seeing and reading, that are currently addressed with ML techniques, are usually handled by human's system 1. However, system 1 exploits its ability to grasp basic notions of causality to build such models, and this does not seem to be present, at least for now, in ML. Another system 1 trait that is lacking in ML is common-sense reasoning (that of course is also related to causality). On the other hand, system 2's capability to solve complex problems seems to be related to AI techniques based on logic, search, optimization, and planning, and employing explicit knowledge, symbols, and high-level concepts.

However, as noted earlier, these two broad sets of capabilities are usually symbiotic, which supports hybrid neuro-symbolic approaches in AI.

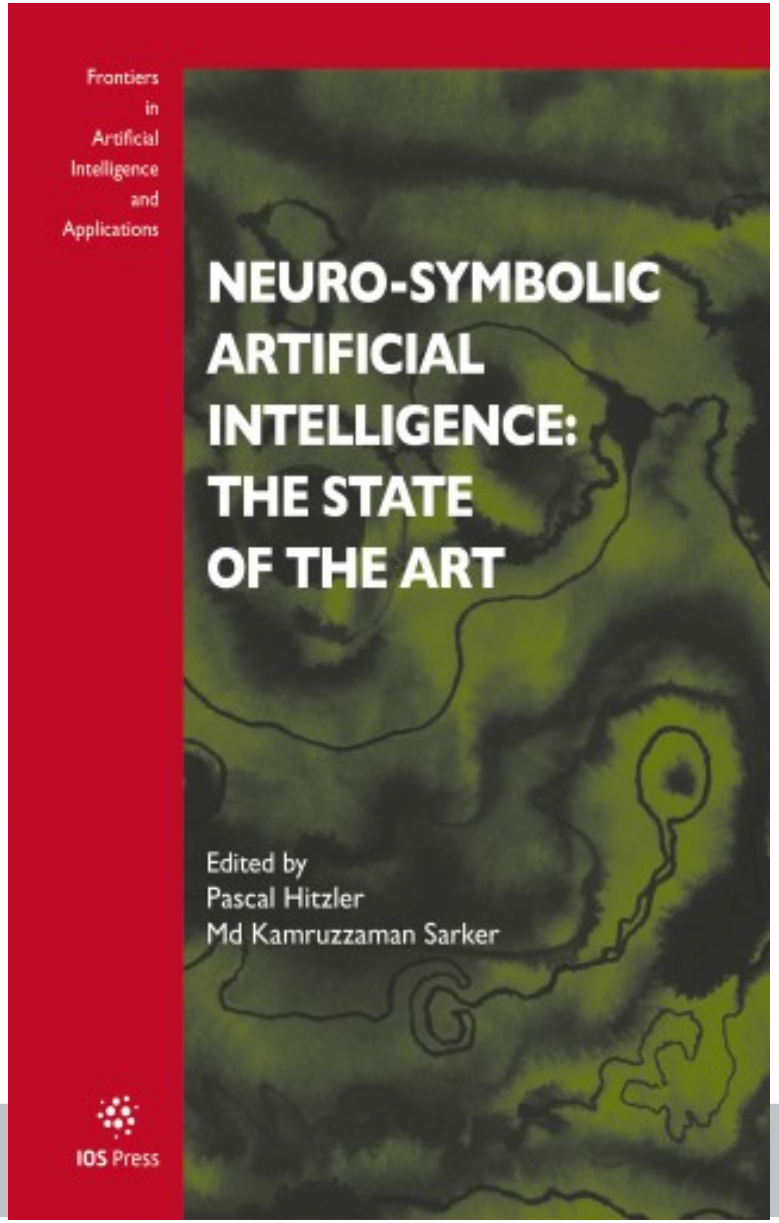
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Neurosymbolic AI
Combining neural networks with symbolic representations might make them more versatile and dependable.
By Don Monroe
Posted Oct 1 2022

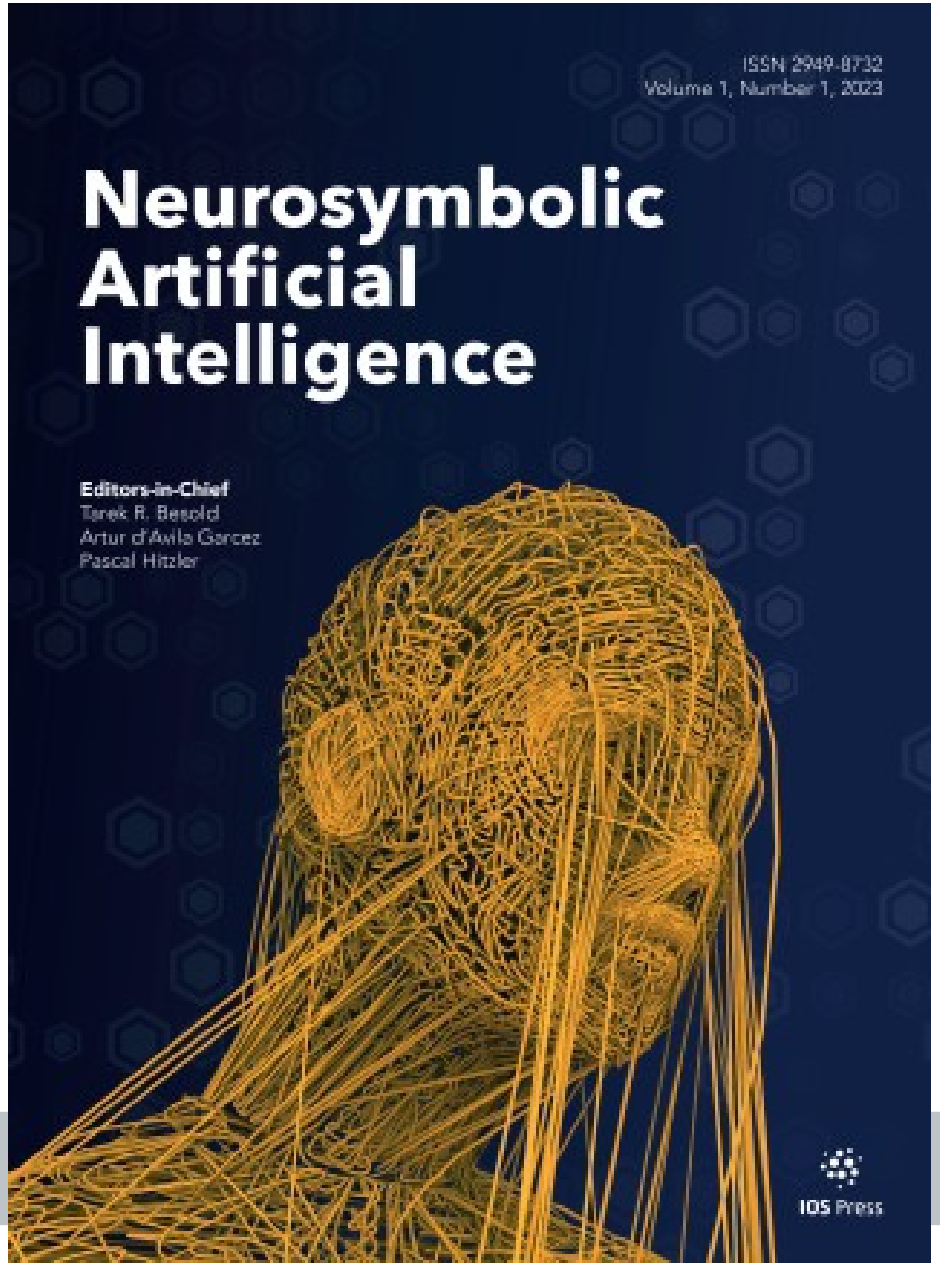
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Neuro-symbolic AI
We see Neuro-symbolic AI as a pathway to achieve artificial general intelligence. By augmenting and combining the strengths of statistical AI, like machine learning, with the capabilities of human-like symbolic knowledge and reasoning, we're aiming to create a revolution in AI, rather than an evolution.

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Frontiers in Artificial Intelligence and Applications
NEURO-SYMBOLIC ARTIFICIAL INTELLIGENCE: THE STATE OF THE ART
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ISSN 2949-8732
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The **Symbolic and Distributed AI** course is about the **Symbolic** component of **Neurosymbolic AI**. It exploits Symbolic AI in the context of **Distributed AI Systems**, where (symbolic, but not only) intelligent software agents distributed across different machines operate autonomously, to coordinate/collaborate in order to achieve some common goal, or to compete to access shared resources. A system consisting of these distributed agents, as well as their environment and the organizations they are part of, is named **Multiagent System** (MAS).

And hence

Welcome on board, and let’s have fun with S&DAI!